# Patient-Level Differential Privacy in Hierarchical Federated Learning for Multimodal Medical IoT

Samartha H V
*Manipal Institute of Technology Bengaluru*
*Manipal Academy of Higher Education*
Manipal, India
samartha1.mitblr2025@learner.manipal.edu

Shreyas J
*Manipal Institute of Technology Bengaluru*
*Manipal Academy of Higher Education*
Manipal, India
shreyas.j@manipal.edu

Udayaprasad P K
*Manipal Institute of Technology Bengaluru*
*Manipal Academy of Higher Education*
Manipal, India
udayaprasad.mitblr2024@learner.manipal.edu

Shreelakshmi Yadav N
*Manipal Institute of Technology Bengaluru*
*Manipal Academy of Higher Education*
Manipal, India
shreelakshmi.mitblr2024@learner.manipal.edu

Sowmya T Rao
*Manipal Institute of Technology Bengaluru*
*Manipal Academy of Higher Education*
Manipal, India
sowmya.mitblr2024@learner.manipal.edu

***Abstract*—Hierarchical federated learning keeps Internet-of-Things data local, but its differential-privacy guarantee protects single records, whereas clinical deployments must protect patients, each with many records. This paper proposes PP-HFM, a client–edge–cloud framework whose privacy unit is the patient. Each client clips a patient's summed gradient, calibrates noise to its population and accounts for every hierarchical release. The guarantee is proved for the training transcript and holds end to end with public encoders and standardization constants (eICU). PP-HFM is evaluated on VitalDB, an eICU cohort of $187$ hospitals and, at record level, a wearable corpus. At a training budget of at most $\varepsilon = 0.9$ ($1.0$ with an evaluated conformal release; $0.766$ at record level on WEAR), it reduces uplink traffic by $90.8$–$96.1\,\%$ and modeled energy per round by $72.3$–$79.6\,\%$; $95$th-percentile inference latency on a workstation CPU is at most $2.3$ ms. On VitalDB, partitioned by person, the guarantee costs no utility (area under the ROC curve, AUC, $0.832$ against $0.820$ non-private); on eICU it costs $3.1$ points (inconclusive against the two-point target) and $3.9$ end to end, which misses it; non-private HED-FL is more accurate ($0.825$ against $0.794$). PP-HFM uploads $2.2$–$4.0$ times less than re-implemented HED-FL and FL2DP and shows lower measured leakage than both in the VitalDB audit. On VitalDB and WEAR, pretrained initialization, not head-only noise, makes a strict budget survivable; the cost of the patient unit follows $A(1 - 1/k)$ in records per patient $k$. A VitalDB membership-inference audit with a positive control finds record-level training above its difficulty floor (likelihood-ratio attack AUC $0.576$) and patient-level training at it ($0.503$).**

***Index Terms*—Hierarchical federated learning, Internet of Things, differential privacy, user-level privacy, membership inference, multimodal learning, edge computing, clinical decision support.**

## I. Introduction

THE Internet of Things (IoT) is increasingly used to monitor people continuously: in the operating room, in the intensive care unit and through wearable sensors. Much of the value of these deployments comes from combining different kinds of data. In an operating room, monitors record physiological waveforms while the hospital record holds the patient's clinical history; in a wearable installation, inertial sensors record motion while a camera records what the wearer sees. A model that sees both streams can recognize states that neither reveals alone.

Such data cannot simply be collected in one place. Clinical records are governed by consent and by data-protection law, and continuous multimodal sensing produces volumes that are costly to move. Federated learning (FL) [1] addresses this by training a shared model where the data reside and exchanging only model updates. Hierarchical FL [2], [3] adds an edge tier between the clients and the cloud, so that most aggregation happens close to the data and the wide-area link is used only once every several rounds, which reduces both traffic and energy.

Keeping the data local, however, does not by itself keep it private. Model updates are computed from the records, and an adversary can often infer from a released model whether a particular record was used in training [4]. Differential privacy (DP) gives a formal guarantee against such inference. Its standard training algorithm, differentially private stochastic gradient descent (DP-SGD) [5], bounds (clips) the influence of each training record and adds calibrated noise, so the guarantee protects whatever unit is clipped. In a sensing deployment that unit is a short window of signal, not a person, and every patient contributes many windows. A record-level guarantee therefore answers whether a particular window was used, whereas a clinical deployment must answer whether a particular *patient* was. Group privacy [6] relates the two, but at the group sizes found in our data it turns a nominal $\varepsilon = 1$ into $\varepsilon \approx 8$ with $\delta \approx 0.09$, which supports no clinical claim.

Existing work has addressed communication efficiency, privacy and multimodal learning largely separately. The review in Section II identifies four gaps.

1) *No patient-level guarantee across a hierarchy.* Hierarchical designs [3], [7], [8] reduce communication and energy but provide no formal guarantee, and private designs [9], [10] provide one but do not account for how an edge tier changes each client's privacy expenditure. Recent systems join a hierarchy with privacy [11], [12] or privacy with multimodal clinical data [13], [14]; to our knowledge, none states its guarantee over an individual whose records are grouped before clipping.
2) *An unmeasured cost.* The utility cost of protecting individuals rather than records has not been measured on multimodal data paired at the source, nor related to the

number of records each individual contributes.

3) *An implicit condition.* Private training of a small head on frozen pretrained features [15], [16] is valid end to end only if the features were pretrained on a population disjoint from the protected one. This condition is usually left implicit, and it is not known whether the benefit comes from the pretraining or from privatizing fewer parameters.
4) *Audits at the wrong unit.* Guarantees are usually reported from mechanism parameters alone and, where audited, audited per record, which is exactly what a record-level guarantee already protects.

This paper proposes PP-HFM (Patient-Private Hierarchical Federated Multimodal learning), a client–edge–cloud framework in which the privacy unit is the patient by construction. Each client sums the gradients of a patient's records before clipping, so that the patient as a whole has bounded influence, and calibrates its noise to its own patient population. The main contributions are as follows.

- *Patient-level privacy with complete accounting (gap 1).* A patient-level private training procedure for hierarchical multimodal FL, with per-client noise calibration, a proof of its guarantee (Theorem 1), and a ledger of every release, edge tier included, that states which configurations are covered end to end.
- *A law for the cost of the unit (gap 2).* A scaling hypothesis, motivated by excess-risk bounds (Proposition 1), under which the utility gap between record- and patient-level training grows as $A(1 - 1/k)$ with the number of records per individual $k$ and vanishes at $k = 1$ (Corollary 1), tested out of sample and on an external corpus.
- *What makes a strict budget affordable (gap 3).* A three-arm design showing that pretrained initialization, not the smaller privatized subset, accounts for robustness at small $\varepsilon$.
- *System validation and an individual-level audit (gaps 1–4).* An evaluation on three multimodal corpora paired at the source, with up to 187 hospital clients; uplink traffic measured from the deployed encoding, delivery and round latency simulated at packet level in ns-3, energy modeled in CloudSim Plus and inference latency timed on a workstation CPU; a patient-level membership-inference audit with a positive control; and a comparison with re-implemented HED-FL and FL2DP under identical data, encoders and seeds.

PP-HFM builds on user-level differential privacy [17] and parameter-efficient private fine-tuning [15], [16]; the contributions are the system that composes them across all tiers, its proof, and its evaluation on three multimodal corpora.

The rest of the paper is organized as follows. Section II reviews related work. Section III states the problem, the system and threat models and the research objectives. Section IV presents PP-HFM and its privacy analysis. Section V describes the experimental setup, Section VI reports the results, Section VII discusses implications and limitations, and Section VIII concludes with directions for future work.

## II. Related Work

This section reviews the four lines of work on which PP-HFM draws and states, for each, what it achieves and what it leaves open.

### A. Hierarchical Federated Learning

Liu *et al.* [2] introduced client–edge–cloud federated learning, in which an edge tier performs several rounds of local aggregation before each global round, so that the wide-area link no longer limits how often clients update. HED-FL [3] clusters devices dynamically at the edge to extend the lifetime of energy-constrained installations. HCEF [7] adapts computation frequency and update compression to device heterogeneity. Mhaisen *et al.* [8] assign clients to edge aggregators using local data statistics, which reduces divergence when client distributions differ. These works establish the communication and energy benefits that PP-HFM adopts. None of them, however, provides a formal privacy guarantee, and the assignment of [8] requires clients to disclose their label statistics, which is itself a release of protected information.

### B. Differential Privacy in Federated Learning

DP-SGD [5] clips the gradient of each record and adds Gaussian noise; its privacy cost is tracked by Rényi [18], [19] or privacy-loss-distribution [20] accountants within the framework of [6]. FL2DP [9] combines noise with shuffling to strengthen the guarantee, and RoPPFL [10] pairs local differential privacy with robust aggregation at the edge. Neither accounts for the edge tier, and neither examines how the tier changes the number of steps, and hence the noise, of each client. User-level differential privacy [17] bounds a user's entire contribution and is the mechanism closest to ours; its rates under heterogeneous contribution are analyzed in [21], but it has not been applied across an edge tier or to multimodal clinical records.

Recent work brings privacy into hierarchical and multimodal settings. PCRHFeL [11] chains sparsification, Wasserstein differential privacy and robust aggregation in a client–edge–cloud hierarchy. Fulcrum [12] allocates noise per participant according to its exposure within a regional aggregate, a calibration close in spirit to the per-client multipliers of PP-HFM, but defined for a participant's class distribution rather than an individual's records. Hasan *et al.* [13] add noise to the local updates of an EHR–ECG fusion model, and MM-PFL-ADP [14] gives client-level guarantees for multimodal imaging. To our knowledge, none of these groups an individual's records before clipping or measures the cost of that unit, which is largest exactly in clinical monitoring, where every participant contributes many records.

### C. Parameter-Efficient Private Adaptation

Training privately only a compact subset of parameters above frozen pretrained features is markedly more robust at small $\varepsilon$ than end-to-end private training [15], [16]. The approach, however, is valid end to end only if the pretraining corpus is disjoint from the protected population. This condition

is rarely stated and its cost rarely measured, and whether the robustness comes from the initialization or from the reduced dimension of the privatized subset has not been separated.

### D. Membership Inference as an Audit

Shadow-model attacks [4] and loss thresholding [22] measure what an adversary achieves in practice, and calibrating for per-example difficulty [23], as in the likelihood-ratio attack (LiRA) [24], makes attacks substantially stronger. Audits, however, are almost always performed per record, and null results are rarely accompanied by a positive control showing that the attack could have detected leakage.

### E. Multimodal and Clinical Federated Learning

Federated learning is well motivated in medicine, where governance rather than data volume prevents pooling [25], [26]. Many multimodal benchmarks pair modalities by a construction rule, so an apparent fusion gain may reflect the rule rather than the data; corpora whose modalities come from the same individual and are aligned at capture [27] or synchronized in hardware [28] avoid this. Conformal prediction [29], [30] adds distribution-free coverage, but its calibration quantile is a statistic of protected data whose private release [31], [32] is costly at small cohorts, as Section VI-B shows. Inference latency at the edge is studied mainly outside federated learning [33], [34].

### F. Summary

Table I compares representative systems, including the unit each guarantee protects and the scope of its accounting. Hierarchical systems lack a formal guarantee, private federated systems protect records or clients rather than individuals, user-level DP has not been applied across a hierarchy, and no system audits its guarantee at the individual level; these are the gaps listed in Section I. Among the works compared, PP-HFM is the only one that combines a hierarchy with a guarantee whose unit is the individual, accounts for every release of the edge tier, and audits the guarantee at the individual level.

TABLE I
COMPARISON WITH REPRESENTATIVE PRIOR SYSTEMS

| Work | H | DP | Unit | Scope | MM | Aud. | Lat. |
|---|---|---|---|---|---|---|---|
| HED-FL [3] | ✓ | – | – | – | – | – | – |
| HCEF [7] | ✓ | – | – | – | – | – | – |
| Mhaisen *et al.* [8] | ✓ | – | – | – | – | – | – |
| FL2DP [9] | – | ✓ | n.s. | F | – | – | – |
| RoPPFL [10] | ✓ | ✓ | U | X | – | – | – |
| PCRHFeL [11] | ✓ | $*$ | n.s. | n.s. | – | – | – |
| Fulcrum [12] | ✓ | ✓ | C | n.s. | – | – | – |
| Hasan *et al.* [13] | – | ✓ | U | F | ✓ | – | – |
| MM-PFL-ADP [14] | – | ✓ | C | F | ✓ | – | – |
| User-level DP [17] | – | ✓ | I | F | – | – | – |
| PP-HFM (this work) | ✓ | ✓ | I | A | ✓ | ✓ | ✓ |

H: hierarchical aggregation; DP: formal guarantee ($*$: Wasserstein differential privacy). Unit: what two neighboring datasets differ by: U, one client update (local DP); C, one client; I, all records of one individual (user or patient). Scope of the privacy accounting: A, every tier, edge included; X, edge tier outside the accounting; F, flat training (no edge tier). MM: multimodal records; Aud.: individual-level audit; Lat.: inference latency reported; n.s.: not specified in the cited work; –: not applicable or not reported.

## III. Problem Statement and System Model

### A. Problem Statement

Consider a medical IoT deployment in which several institutions, such as hospitals, surgical specialties or wearable installations, each hold multimodal recordings of their own patients and wish to train a shared prediction model. Every patient contributes many records, and no record may leave its institution. The problem is to train the model through a client–edge–cloud hierarchy such that

1) the participation of any single patient, with all of their records, is protected by a formal $(\varepsilon, \delta)$ guarantee that covers every message the system releases, edge tier included;
2) the model remains close in utility to an identical non-private system; and
3) the system remains deployable on IoT infrastructure, with low uplink traffic, reliable delivery, low inference latency and low energy.

These requirements interact. Protecting patients instead of records reduces the number of protected units and therefore raises the relative noise. Clients partitioned by institution differ greatly in size, so a single noise level fitted to the smallest client degrades every larger one. And an edge tier changes how many optimization steps each client takes, and hence how much of its budget it spends. The remainder of this section formalizes the setting; Section IV addresses each difficulty in turn.

### B. System Model

A deployment has $K$ clients, such as hospitals, specialties or wearable installations, assigned by a fixed public map $e(\cdot)$ to $E$ edge aggregators, which report to one cloud aggregator. Client $k$ holds a dataset $\mathcal{D}_k$ of $n_k$ records. Each record pairs two modalities of the same individual over the same interval: modality A is a multichannel time series and modality B a static or summarized feature vector. The records of client $k$ are partitioned by individual into the patient set $\mathcal{P}_k$; patient $p$ contributes the records $W_p$, with $k_p = |W_p|$. A patient's records never span two clients or two evaluation splits; this holds by construction on eICU and on the person-level VitalDB partition (Section V). The model $f_\theta$, with $\theta = (\theta_{\text{enc}}, \theta_{\text{head}})$, comprises two modality encoders and a fusion head, and the learning objective is the weighted empirical risk

$$\min_{\theta} \; F(\theta) = \sum_{k=1}^{K} \frac{n_k}{n} \, \frac{1}{n_k} \sum_{(x,y)\in\mathcal{D}_k} \ell\big(f_\theta(x), y\big), \tag{1}$$

with $n = \sum_k n_k$ and $\ell$ the unweighted cross-entropy; class weights computed from training labels would be a further unprotected release. Training runs for $R$ cloud rounds of $\tau_e$ edge rounds each.

Table II summarizes the notation.

TABLE II
NOTATION.

| Symbol | Meaning |
|---|---|
| $K$, $E$ | number of clients, of edge aggregators |
| $\mathcal{P}_k$, $W_p$ | patients of client $k$; records of patient $p$ |
| $k_p = \lvert W_p \rvert$ | records contributed by patient $p$ |
| $R$, $\tau_e$ | cloud rounds; edge rounds per cloud round |
| $B$, $q_k$ | expected patients per step; sampling rate |
| $C$, $\sigma_k$ | clipping norm; noise multiplier of client $k$ |
| $T_k$ | optimization steps of client $k$ |
| $(\varepsilon, \delta)$ | privacy budget |

### C. Threat Model and Privacy Goal

The adversary observes everything that leaves a client: every encoded update, and every edge and cloud model. It may also know all records except those of a target individual. Edge and cloud aggregators are honest-but-curious and are treated as part of the adversary. The goal is $(\varepsilon, \delta)$-differential privacy [6] under the *patient-level* neighbouring relation: two datasets are neighbours if one is obtained from the other by adding or removing *all* records of a single patient. A record-level guarantee, which removes one record, does not meet this goal when $k_p > 1$. Public quantities are the architecture, the pretrained encoder parameters, all hyperparameters, the edge map, the declared client sizes $|\mathcal{P}_k|$ and $n_k$, and the input standardization constants, which are fixed before training and are public only when fitted on the disjoint pool (Table III). The validation cohort, used to select the reported checkpoint, is outside the guarantee, as is conventional [5]; Section IV-E lists every release.

### D. Research Objectives

To make the three requirements testable, seven quantitative targets were fixed before any experiment was run:

- O1: reduce uplink traffic by at least $50\,\%$ relative to centralized training;
- O1b: reduce the latency of a training round relative to centralized training;
- O2a: keep the privacy budget at $\varepsilon \leq 1$ for every protected individual;
- O2b: lose at most two points of utility relative to an identical non-private system;
- O3a: keep the 95th-percentile inference latency below $100$ ms;
- O3b: deliver more than $99\,\%$ of updates; and
- O4: save at least $20\,\%$ of energy per training round.

Section VI-A reports each target on every corpus.

## IV. The PP-HFM Framework

### A. Overview

Fig. 1 shows the architecture of PP-HFM. It has three tiers: clients, which hold the patients' records; edge aggregators, each serving a group of clients; and a cloud aggregator. Each client holds patient-grouped multimodal records that never leave the institution. A training round proceeds in four stages, each described in one of the following subsections.

1) *Local encoding.* The client passes each record through two frozen, pretrained encoders, one per modality.
2) *Patient-level private training* (Section IV-B). The client trains only the fusion head, grouping each patient's gradients before clipping and adding noise.
3) *Per-client noise calibration* (Section IV-C). Before training, each client sets its noise level from its own population size, so that every patient receives the same budget.
4) *Hierarchical aggregation* (Section IV-D). The client sends a sparsified, quantized head update to its edge; each edge averages its clients' updates for $\tau_e$ rounds and then reports to the cloud, which averages across edges and broadcasts the global model.

Section IV-E then proves the guarantee and Section IV-F describes how it is audited. The encoder architecture and the averaging rule are adopted from established work; the contribution lies in the unit at which privacy is enforced and in the accounting of the reported budget.

*Frozen multimodal backbone.* As shown in Fig. 2, the temporal encoder applies three parallel one-dimensional convolutions with different kernel widths, a strided convolution, a further convolution with max pooling, a two-layer bidirectional recurrent layer and attention pooling; the static encoder is a two-layer perceptron. The two embeddings are concatenated and mapped to class logits by the fusion head which holds $559{,}874$ of the $1{,}864{,}227$ parameters ($30.0\,\%$). Every normalization layer is a group or layer normalization: batch normalization mixes statistics across a batch, which would make the per-patient gradient ill-defined and the accounting invalid. The encoders are pretrained centrally and frozen. Two configurations are distinguished. In the *disjoint* configuration, a set of individuals is reserved before federated training, verified disjoint from every client's splits, and used alone for pretraining; with standardization constants also fitted on that pool, this configuration satisfies the guarantee end to end. In the *oracle* configuration, the encoders are pretrained on the pooled client training data; it does not satisfy the guarantee end to end and is used only to isolate the behavior of the private head, so its utilities are upper bounds.

### B. Patient-Level Private Training

This stage ensures that each patient, and not each record, has bounded influence on the model. The mechanism fixes the unit at the point of clipping rather than recovering it afterwards; Fig. 3(a) contrasts the two. At each optimization step, client $k$ draws a Poisson sample $S \subseteq \mathcal{P}_k$, including each patient independently with probability

$$q_k = \min\big(1,\ B/|\mathcal{P}_k|\big), \tag{2}$$

where $B$ is the expected number of patients per step. For each sampled patient, the gradients of all of that patient's records are summed and the sum is clipped,

$$g_p = \sum_{(x,y)\in W_p} \nabla_{\theta_{\mathrm{head}}}\, \ell\big(f_\theta(x), y\big), \qquad \bar{g}_p = g_p \min\Big(1, \frac{C}{\|g_p\|_2}\Big). \tag{3}$$

Because the bound is applied to the sum, patient $p$ changes the aggregate by at most $C$ irrespective of $k_p$; clipping each

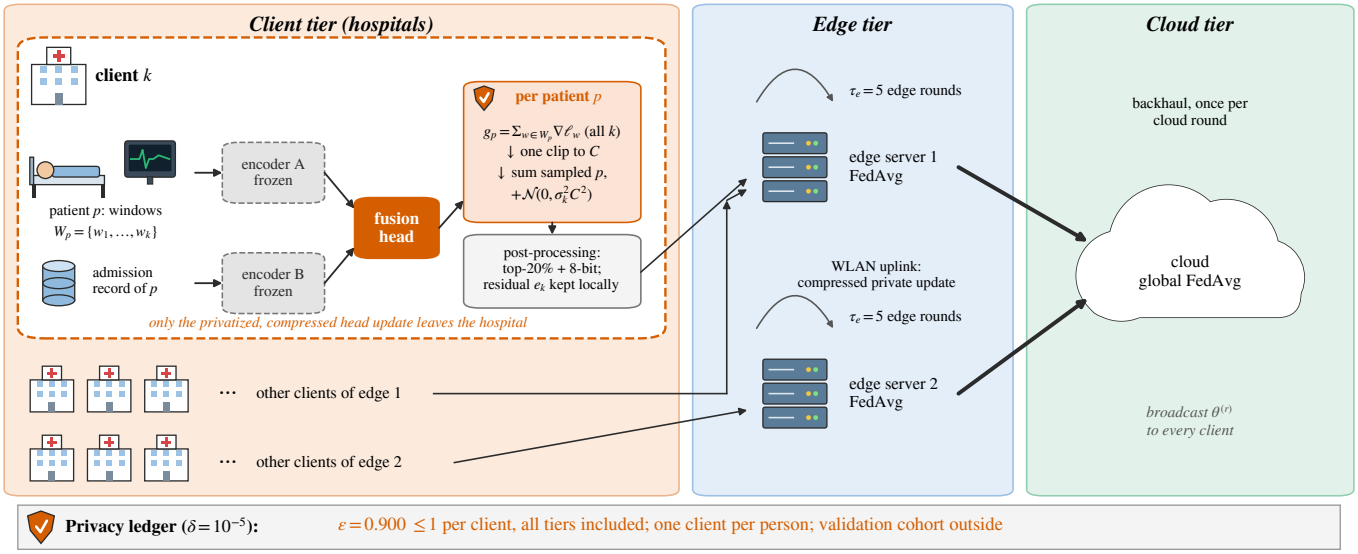



Fig. 1. Architecture of PP-HFM. Clients train only the fusion head under patient-level DP; edges and the cloud average the noisy, compressed updates.

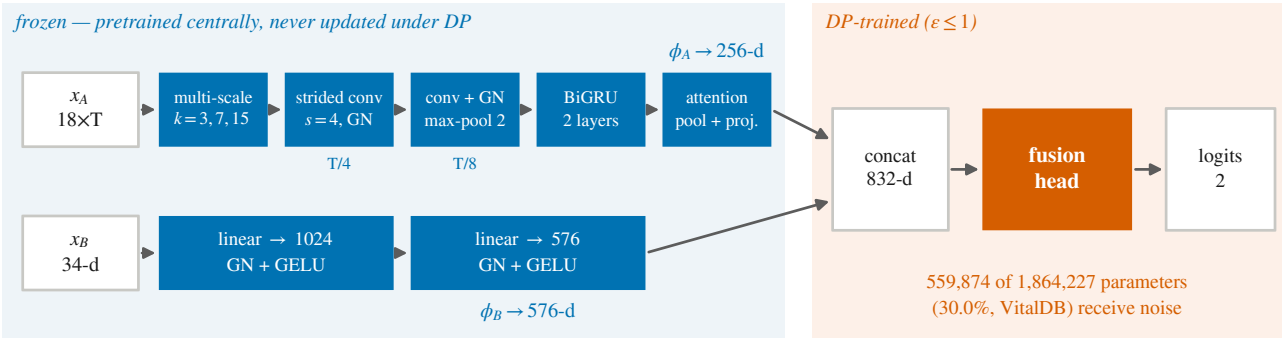



Fig. 2. Encoder–fusion backbone (VitalDB). Encoders are pretrained and frozen; only the fusion head (559,874 parameters) is privatized.

record would admit $k_p C$. Gaussian noise scaled to the same norm is added once per step, and the result is divided by the *expected* batch size,

$$\tilde{g} = \frac{1}{q_k |\mathcal{P}_k|} \Big( \sum_{p \in S} \bar{g}_p + \mathcal{N}\big(0, \sigma_k^2 C^2 I\big) \Big), \tag{4}$$

after which $\theta_{\text{head}} \leftarrow \theta_{\text{head}} - \eta \tilde{g}$; the implementation passes $\tilde{g}$ to Adam instead, which acts only on the released gradient and is post-processing. The divisor must not be the realized $|S|$: the noise norm reveals the divisor, $|S|$ depends on whether a given patient is present, and normalizing by it releases information the accountant does not cover. For the same reason a step whose sample is empty still releases the noise. Algorithm 1 states the procedure. For compute, at most 64 records of a patient enter $g_p$; the rule is fixed and public, so the guarantee is unaffected, and it truncates 4.1 % of eICU patients (3.5 % of training windows) and no VitalDB patient.

*Objective solved.* Without clipping, the expected step is $\mathbb{E}[\tilde{g}] = |\mathcal{P}_k|^{-1} \sum_p g_p = \bar{k}_k \nabla F_k(\theta)$, where $F_k$ is the record mean of (1) on client $k$ and $\bar{k}_k = n_k / |\mathcal{P}_k|$. A local pass takes $|\mathcal{P}_k|/B$ such steps, $\bar{k}_k$ times fewer than a record-level pass with $B$ records per step, so to first order in $\eta$ both displace client $k$ by $\eta\,(n_k/B)\,\nabla F_k$: before clipping the patient-level mechanism targets (1) with unchanged client weights, in fewer and larger steps. The argument is for the plain gradient step; Adam rescales coordinates adaptively, so in the implementation it holds only approximately. Clipping is what changes the objective: each patient's summed gradient is bounded as a whole, the surrogate of user-level DP-SGD [17], which down-weights patients whose summed gradient exceeds $C$, typically those with many records. This change belongs to the unit and is part of its measured cost (Section VI-D).

### C. Per-Client Noise Calibration

This stage sets how much noise each client adds. Clients partitioned by institution differ greatly in size, and therefore in sampling rate: a small client samples a large fraction of its patients per step. A single multiplier calibrated for the most demanding client satisfies the budget everywhere but imposes that client's noise on all others; on eICU, a hospital of fifteen patients sampled at $q_k = 1$ would set the noise of all 187

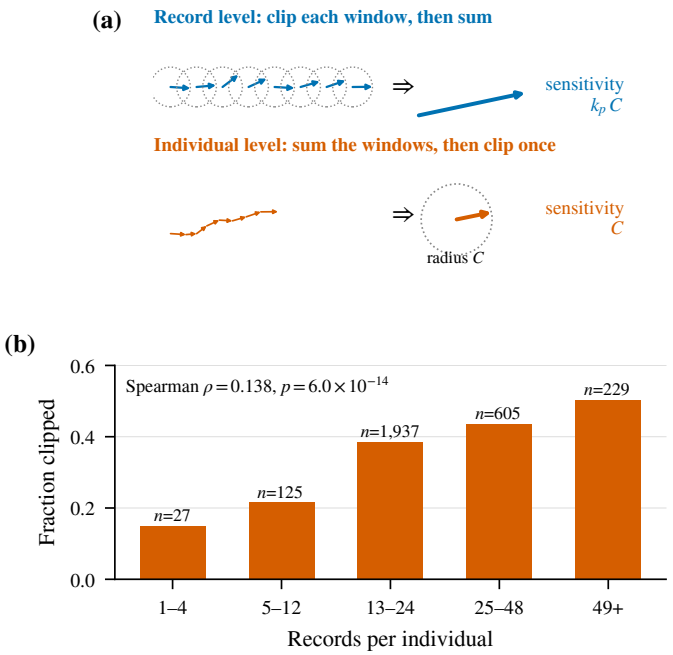



Fig. 3. Unit of the guarantee. (a) Per-window versus per-patient clipping. (b) Fraction of eICU individuals clipped at $C = 1$ against records contributed.

**Algorithm 1** Patient-level private local update at client $k$

**Require:** frozen $\theta_{\text{enc}}$; head $\theta_{\text{head}}$; patients $\mathcal{P}_k$ with records $\{W_p\}$; clip $C$; multiplier $\sigma_k$ from (5); rate $q_k$ from (2); step size $\eta$

**for** each local step **do**
  Sample $S \subseteq \mathcal{P}_k$, each $p$ independently w.p. $q_k$
  **for** $p \in S$ **do**
    $g_p \leftarrow \sum_{(x,y)\in W_p} \nabla_{\theta_{\text{head}}} \ell(f_\theta(x), y)$ ▷ group by patient
    $\bar{g}_p \leftarrow g_p \cdot \min(1, C/\|g_p\|_2)$ ▷ bound the sum
  **end for**
  $\tilde{g} \leftarrow \big(\sum_{p\in S} \bar{g}_p + \mathcal{N}(0, \sigma_k^2 C^2 I)\big)/(q_k|\mathcal{P}_k|)$ ▷ also if $S = \emptyset$
  $\theta_{\text{head}} \leftarrow \theta_{\text{head}} - \eta\, \tilde{g}$
**end for**
**return** $\theta_{\text{head}}$

clients. When each patient is held by exactly one client (A1), the clients' mechanisms act on disjoint individuals, and by parallel composition a patient's guarantee is that of the client holding them; individuals held by several clients are treated in the scope statement of Section IV-B. Each client therefore uses the smallest multiplier that meets the training budget for its own sampling rate and step count,

$$\sigma_k = \min\{\sigma : \ \varepsilon_k(\sigma, q_k, T_k) \le \varepsilon\}, \qquad T_k = R\,\tau_e \max\big(1, \lfloor |\mathcal{P}_k|/B \rfloor\big), \tag{5}$$

where $\varepsilon_k$ is computed by the privacy loss distribution accountant [20], a tight upper bound for the subsampled Gaussian mechanism, and $T_k$ must equal the number of steps the client actually executes. Since $\sigma_k$ depends only on the declared, public $|\mathcal{P}_k|$ and $T_k$, using a different multiplier per client releases nothing further. Arm-versus-arm comparisons in Section VI

**Algorithm 2** PP-HFM cloud round $r$

**Require:** global model $\theta^{(r-1)}$; edge map $e(\cdot)$; edge rounds $\tau_e$

$\theta_e^{(r,0)} \leftarrow \theta^{(r-1)}$ for every edge $e$
**for** $s = 1$ **to** $\tau_e$ **do**
  **for** each client $k$ **in parallel do**
    $\theta_k^{(r,s)} \leftarrow$ Algorithm 1 from $\theta_{e(k)}^{(r,s-1)}$
    transmit $Q\big(\text{top-}\kappa(\theta_k^{(r,s)} - \theta_{e(k)}^{(r,s-1)})\big)$
  **end for**
  each edge aggregates by (6) (left)
**end for**
cloud aggregates by (6) (right) and broadcasts $\theta^{(r)}$
**return** $\theta^{(r)}$

use the single conservative multiplier $\sigma^\star = \max_k \sigma_k$ with Rényi accounting [18], [19], which holds every arm to the same noise.

### D. Hierarchical Aggregation and Update Encoding

This stage combines the clients' private updates while keeping the uplink small. Within a cloud round $r$, each client initializes from its edge model, performs one private local pass, and transmits its encoded update. Edge $e$ and the cloud compute

$$\theta_e^{(r,s)} = \sum_{k:e(k)=e} \frac{n_k}{N_e}\, \theta_k^{(r,s)}, \qquad \theta^{(r)} = \sum_{e=1}^{E} \frac{N_e}{n}\, \theta_e^{(r,\tau_e)}, \tag{6}$$

with $N_e = \sum_{k:e(k)=e} n_k$ (Algorithm 2). The weights are public constants; weights computed from the records drawn in a round would depend on which patients were sampled and would leave the client unprotected. With $\tau_e = 1$ the edge tier collapses into conventional single-tier federated averaging. Each update $\Delta$ of dimension $d$ is sparsified to its $\kappa d$ largest-magnitude coordinates and the retained values are uniformly quantized to $b$ bits,

$$Q(v) = v_{\min} + \text{round}\Big(\frac{v - v_{\min}}{s}\Big)\, s, \qquad s = \frac{v_{\max} - v_{\min}}{2^b - 1}, \tag{7}$$

with the suppressed residual accumulated locally and added to the next update. Both operations act on the already-perturbed head and are post-processing.

### E. Privacy Analysis

This subsection proves that, under three conditions, the procedure above protects every training patient, states what the protection costs, and lists every quantity the system releases. Three conditions are assumed. (A1) Each patient's records lie in the training split of exactly one client, and the training cohort is disjoint in patients from the validation cohort, which serves checkpoint selection. (A2) The architecture, the encoder parameters, which are pretrained on a patient-disjoint pool, all hyperparameters, the edge map, the standardization constants and the declared client sizes $|\mathcal{P}_k|$ and $n_k$ are public and fixed before training; $q_k$, $T_k$, $\sigma_k$ and the weights of (6) are computed

from them and never from the data. (A3) The forward pass of a record depends on that record alone.

*Theorem 1 (Patient-level guarantee):* Under (A1)–(A3), let each $\sigma_k$ satisfy (5). The transcript of a PP-HFM run, consisting of every encoded client update, every edge and cloud model and the model selected on the validation cohort, is $(\varepsilon, \delta)$-differentially private with respect to each training patient, under adding or removing all records of one patient with the parameters of (A2) held fixed.

*Proof:* Fix a training patient $p$, held by client $k$ by (A1). *One step.* Given the model entering a step, the clipped gradients $\bar{g}_{p'}$, $p' \neq p$, are unaffected by $p$ by (A3), and $\|\bar{g}_p\|_2 \leq C$ by (3). The noisy sum in (4) is therefore a Poisson-subsampled Gaussian mechanism with rate $q_k$, noise $\sigma_k C$ and $\ell_2$-sensitivity $C$, whose privacy loss is bounded in [19], [20]; division by the public constant $q_k|\mathcal{P}_k|$ and the update are post-processing. *Composition.* Each step takes as input earlier outputs, other clients' messages and public quantities, so the $T_k$ steps compose adaptively, and (5) ensures $\varepsilon_k \leq \varepsilon$ at $\delta$. *Other clients.* By (A1) no other client accesses the records of $p$; its outputs are randomized functions of its own data and of already-released messages and contribute no privacy loss with respect to $p$. Since $\sigma_{k'}$ depends only on public quantities, the per-client multipliers release nothing. *Hierarchy.* Sparsification with a local residual, quantization, averaging with the public weights of (6) and broadcast are functions of released outputs and public quantities, and selection on the validation cohort uses data disjoint from $p$; all are post-processing. ∎

*Adjacency and public sizes.* Adding or removing a patient changes the realized cohort, but not the mechanism: the rate, step count, multiplier, divisor and aggregation weights are fixed by the declared sizes before training, so both neighbouring datasets are processed by the same mechanism. This is the setting in which the Poisson-subsampled Gaussian is analyzed [19], [20]; it needs no knowledge of the realized dataset and its guarantee holds for a dataset of any size. The declared sizes are themselves released. In the experiments they equal the realized cohort sizes, which institutions typically publish; where a size is sensitive, it can be replaced by a coarse public value or released by a private count at a small additional budget, and the theorem holds for whichever declared values are used. *Standardization.* The theorem holds with the standardization constants fixed, like the declared sizes; with the constants recomputed after a patient is removed, every other record would be represented differently. In the experiments the constants are fitted once on the training cohort, so their release is outside the guarantee (Table III). Its size was measured on eICU, where the fit can be reproduced exactly: removing one individual moves any mean by at most $0.0017$ standard deviations and, for $99\,\%$ of individuals, any standard deviation by at most $0.02\,\%$, but the most extreme outlier moves one channel's standard deviation by $5.2\,\%$. Constants fitted on the disjoint pretraining pool alone differ from those used by at most $0.026$ standard deviations in mean and $4.4\,\%$ in scale. On eICU the headline is therefore repeated with constants fitted on the disjoint pool and encoders pretrained in those units (Section VI-B); there every input to the private head is either public or covered by the theorem.

TABLE III
PRIVACY LEDGER OF THE HEADLINE CONFIGURATION: EVERY RELEASE AND THE POPULATION IT TOUCHES. TABLE IV GIVES THE OTHER CONFIGURATIONS.

| Release | Population | Protection |
|---|---|---|
| Client updates, edge/cloud models | training | $(0.9, 10^{-5})$-DP per client |
| Selected checkpoint | validation | none |
| Conformal threshold (evaluated) | validation | $\varepsilon_{\mathrm{conf}} = 0.1$ |
| Standardization (headline runs) | training | none |
| Standardization (eICU rerun) | public pool | public |
| Declared sizes, $q_k$, $T_k$, $\sigma_k$ | – | public by design |
| Encoders (disjoint pool) | external | not required |
| Encoders (oracle) | training | violates (A2) |

Table III lists every release of the headline configuration; the unit comparison and the audit use $\varepsilon = 1.0 + 0.5$ with one worst-client multiplier (Table IV). The protected population is the training cohort; validation patients select the checkpoint without noise and are outside the guarantee, as is conventional [5].

*Corollary 1 (Coincidence of the units):* If every patient contributes one record, Algorithm 1 is DP-SGD with Poisson sampling over records. For equal $(B, T_k, \sigma, C)$, the patient- and record-level mechanisms are then the same mechanism, so any difference between them at one record per patient measures seed variation or an implementation difference, not the privacy unit.

The theorem bounds what the guarantee reveals; the next statement predicts what it costs. Suppose $P$ individuals each contribute exactly $k$ records. The record-level arm minimizes $\frac{1}{Pk}\sum_p \sum_{w \in W_p} \ell_w$, and the patient-level arm minimizes the same objective written as $\frac{1}{P}\sum_p \bar{\ell}_p$ with $\bar{\ell}_p = \frac{1}{k}\sum_{w \in W_p} \ell_w$.

*Proposition 1 (Cost in records per individual):* Let each $\ell_w$ be convex and $L$-Lipschitz on a domain of diameter $D$ in $\mathbb{R}^d$, and let both arms be trained by private gradient descent at the same $(\varepsilon, \delta)$, and let $\Lambda = LD\sqrt{d\log(1/\delta)}/\varepsilon$. (i) The excess empirical risk of the record-level arm is at most $c_{\mathrm{rec}}\Lambda/(Pk)$ and that of the patient-level arm at most $c_{\mathrm{ind}}\Lambda/P$, with constants independent of $k$; both rates are tight in the worst case up to logarithmic factors. (ii) If both arms attain their rate with a common constant $c$, the gap is $A(1 - 1/k)$ with $A = c\Lambda/P$.

*Proof:* For $n$ units with convex $L$-Lipschitz losses, private gradient descent attains excess risk $O(LD\sqrt{d\log(1/\delta)}/(n\varepsilon))$, and the rate is tight up to logarithmic factors [35]. The record-level arm has $n = Pk$ units. In the patient-level arm the unit is $\bar{\ell}_p$, which is $L$-Lipschitz as an average of $L$-Lipschitz losses; clipping the summed gradient at $C$ equals clipping the gradient of $\bar{\ell}_p$ at $C/k$, a rescaling that leaves the guarantee unchanged, so the mechanism is private gradient descent over $n = P$ units. This gives (i), and (ii) follows by subtraction. ∎

Part (ii) rests on an assumption that (i) does not supply: upper bounds with different constants do not determine the attained gap. The form $A(1 - 1/k)$ is therefore a *scaling hypothesis* that the proposition motivates, not a theorem, and it is tested empirically, including out of sample. Since the head is not convex and the reported metric is not excess risk, the hypothesis concerns the *shape* of the cost, not its

magnitude: one free parameter, zero at $k = 1$, and a patient-level arm whose utility does not depend on $k$. It also indicates what would recover the cost: mechanisms that clip around an estimate of each individual's mean gradient attain user-level rates that improve with $k$ [21], whereas a bound fixed around zero cannot.

*Scope.* On eICU and on the person-level VitalDB partition the unit is the person. On the case-level VitalDB partition it is the surgical case: 230 of 5,974 people (3.9 %) underwent more than one operation, and for them the guarantee extends to the person only by group privacy over at most 14 cases, composed across clients for the 68 training persons whose cases lie on two. On eICU, where clients are hospitals, the stays of 2,177 of 74,327 training persons (2.9 %) lie at two to four hospitals, so in the source partition (A1) fails for them. The privacy-cost and end-to-end runs therefore use a partition in which each such person is moved, with all their records, to the hospital holding most of their stays (49,939 training records; likewise 509 validation and 589 test persons); no record is excluded and every split is unchanged. In that partition every training and validation person is held by one hospital, so (A1) holds and the guarantee is $\varepsilon_{\text{train}} = 0.9$ for every training person, verified independently for every client of every run (privacy-loss-distribution bounds $[0.880, 0.900]$; Rényi accountant 0.974), and the evaluated conformal release adds $\varepsilon_{\text{conf}} = 0.1$ once per person. The remaining eICU analyses (system objectives other than O2, scaling grid, clipping rule and prior systems) use the source partition, in which the per-client budget composes across the hospitals that hold a person: with the headline accountant, at most 1.32, 1.64 and 1.92 for the 2,097, 75 and 5 persons at two, three and four hospitals (Rényi: 1.42, 1.77, 2.08), and at most $m\varepsilon$ in general; arms compared with each other share the partition. On WEAR each client is one participant, client-, user- and record-level units coincide at the client, and the mechanism evaluated there is record-level. The guarantee holds end to end only with the disjoint encoders and pool-fitted standardization constants; the oracle encoders violate (A2), and training-fitted constants are an unprotected release. Table IV lists, for each evaluated configuration, the unit, the provenance of the encoders and standardization constants, the accounting and what the guarantee covers. The implementation of the accounting is additionally checked by a suite of 17 regression checks, seven of which cover it: composition of the budget across calls, release and accounting of noise at every step with the expected-batch divisor, one conformal score per calibration patient, and per-client multipliers within the budget. These are implementation-level checks of the accounting configuration; they do not by themselves establish the guarantee.

### F. Empirical Audit

A proof bounds what an adversary can learn; an audit measures what a concrete attack actually achieves. The guarantee is audited by patient-level membership inference. Members are training patients and non-members are held-out patients, scored by the same released model. Two attacks are used: thresholding of a patient's mean loss [22], and LiRA [24]

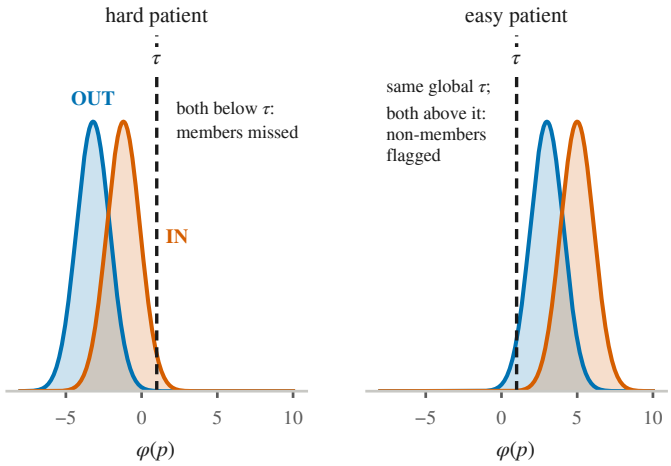



Fig. 4. Per-patient calibration in LiRA (illustrative): a global threshold misclassifies hard and easy patients that per-patient IN/OUT fits separate.

adapted to the patient unit. Sixteen shadow models are trained on independently drawn halves of the patient population, with all records of a patient assigned together. For each patient, the logit-scaled confidence averaged over their records,

$$\varphi(p) = \frac{1}{k_p} \sum_{(x,y)\in W_p} \Big( z_y(x) - \log \sum_{j \neq y} e^{z_j(x)} \Big), \qquad (8)$$

is modeled by Gaussians fitted over the shadows that include and exclude the patient (Fig. 4), and membership is scored by $\Lambda(p) = \log \mathcal{N}(\varphi; \mu_{\text{in}}, \sigma^2_{\text{in}}) - \log \mathcal{N}(\varphi; \mu_{\text{out}}, \sigma^2_{\text{out}})$. A negative result is accepted only if the same procedure detects clearly elevated leakage on a positive control, non-private training of the same head over the same encoders. Attack AUCs are compared with a difficulty floor, the AUC obtained by ranking patients by their mean statistic over the shadows that exclude them, without the target model, because members and non-members may differ in intrinsic difficulty. The audited models are the five-seed record- and patient-level models of the unit comparison (Section VI-B), trained at $\varepsilon = 1$, a slightly weaker budget than the headline, so the audit is conservative.

### G. Deployment and Instrumentation

The trained network is exported to a portable graph and executed at full precision on a CPU provider at the edge, so the wide-area link is not on the inference path. Latency is decomposed into preprocessing, forward pass and post-processing, and is reported at the 95th percentile over 1,000 timed runs after 100 warm-up runs. Communication delay, delivery and traffic are measured in a packet-level network simulation of the three tiers, with a wireless local-area device-to-edge link and a point-to-point backhaul, on every simulated round. Energy is modeled per tier, because a hierarchy redistributes work, and an aggregate figure would hide a saving at one tier offset by a cost at another.

## V. Experimental Setup

### A. Datasets

Three public corpora are used (Table V); Fig. 5 outlines the study. In each, the two modalities come from the same

TABLE IV
SCOPE OF THE GUARANTEE IN EACH EVALUATED CONFIGURATION

| Configuration | Unit | Encoders / standardization | Accounting | Guarantee covers |
|---|---|---|---|---|
| eICU, headline (one hospital per person) | patient | disjoint / training cohort | $\sigma_k$, PLD; $0.9 + 0.1$ | training transcript |
| eICU, pool-fitted constants (one hospital per person) | patient | disjoint / public pool | $\sigma_k$, PLD; $0.9 + 0.1$ | end to end |
| VitalDB, person-level headline | person | oracle / training cohort | $\sigma_k$, PLD; $0.9 + 0.1$ | transcript; (A2) violated |
| VitalDB, disjoint encoders (case-level) | surgical case | disjoint / training cohort | $\sigma_k$, PLD; $0.9 + 0.1$ | transcript, per case |
| VitalDB, cost of the unit and audit (person-level) | person vs record | oracle / training cohort | $\sigma_{\max}$, RDP; $1.0 + 0.5$ | training transcript |
| Scaling grids, clipping rule (VitalDB case-level; eICU) | individual vs record | oracle (VitalDB), disjoint (eICU) / training | $\sigma_{\max}$, RDP; $0.8$–$1.0 + 0.2$–$0.5$ | training transcript‡ |
| eICU, prior systems (source partition) | patient | disjoint / training cohort | $\sigma_k$, PLD; $0.9 + 0.1$ | training transcript‡ |
| WEAR | record (= participant) | oracle / training cohort | record-level; $0.766 + 0.2$ | transcript, per record |

Unit: neighboring datasets differ by all records of one patient, person or case, or by one record; "vs" marks a comparison of two arms. Accounting: $\varepsilon_{\text{train}} + \varepsilon_{\text{conf}}$ at $\delta = 10^{-5}$; $\sigma_k$, per-client multipliers; $\sigma_{\max}$, one worst-client multiplier; PLD, privacy-loss-distribution accountant; RDP, Rényi accountant. The conformal release was evaluated and is not part of the deployed system. Arms compared with each other share one configuration. Other analyses (recipe, ablation, fusion, attribution) are labeled with their partition and mechanism in Section VI. ‡eICU on the source partition: for the 97.1 % of training persons at one hospital; the 2.9 % at two to four hospitals compose the mechanisms of those hospitals ($\varepsilon_{\text{train}} \leq 1.32$–$1.92$ with $\sigma_k$ and PLD).

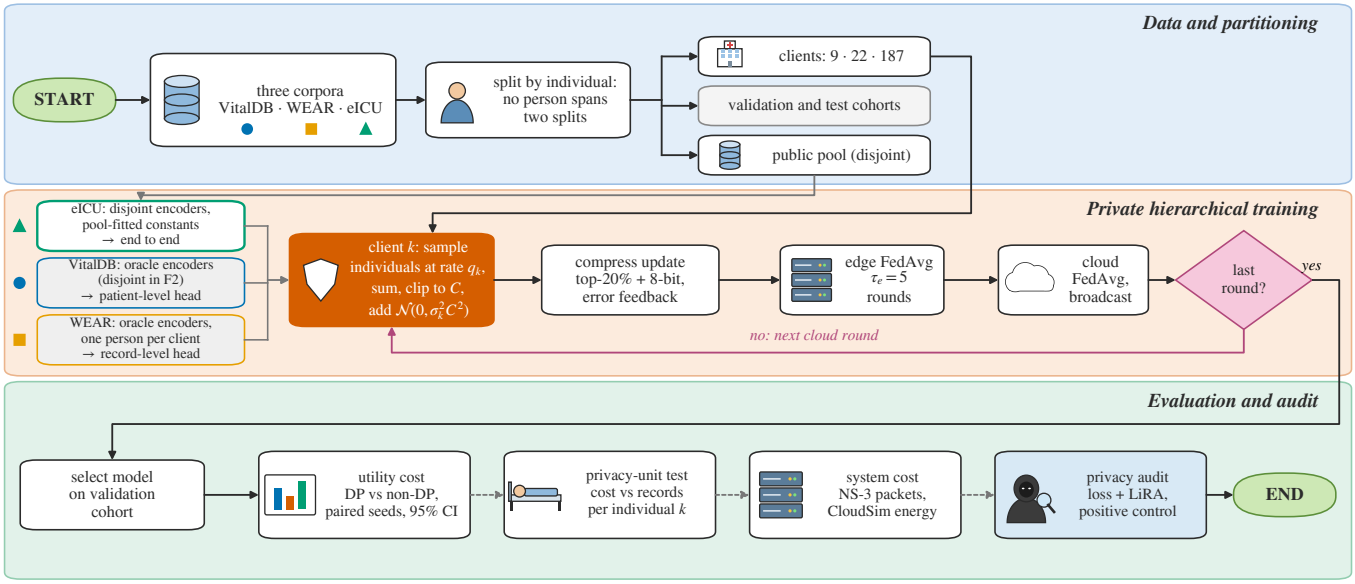



Fig. 5. Study workflow: individual-level partitioning, per-corpus encoder configuration (end to end on eICU only), private hierarchical training and paired evaluation with an attack audit.

individual and are paired at the source rather than by a construction rule, which a claim about fusion requires. They also cover the two regimes that matter for the mechanism: clients holding many individuals (VitalDB, eICU) and a client holding one (WEAR). WEAR is not clinical: it is a wearable IoT workload that tests the system objectives outside medicine, and its privacy results are record-level.

*VitalDB* [27] contains non-cardiac surgical cases. Modality A is ten minutes of nine intraoperative trends at 0.5 Hz with nine binary indicators of whether each was recorded; modality B is a 34-dimensional vector of preoperative clinical and laboratory variables. The target is postoperative intensive-care admission. Clients are the nine surgical specialties, which differ by a factor of almost eight in size and by more than fifteen in outcome prevalence (3.0–50.4 % on the person-level partition; 3.2–50.7 % on the case-level partition of Fig. 6), a heterogeneity that arises from how care is organized rather than from an imposed skew. *WEAR* [28] combines body-worn inertial sensing with egocentric video features synchronized in hardware; each of the 22 clients is one participant. *eICU* [36], distributed through PhysioNet [37] under a credentialed agreement, is the external clinical corpus. Modality A is the first 24 hours after unit admission in 60-minute windows of six vital signs and six observation indicators; modality B is age, sex, admission weight and height and the APACHE score, a clinician-computed summary rather than an independent sensor stream; the target is in-unit mortality. Prediction is made at hour 24: individuals who die within the first 24 hours are excluded, and the APACHE score, computed from the worst values of the first day, is therefore available at prediction time. Clients are the hospitals themselves: of 206, those with fewer than 200 windows or no positive outcome are pooled, giving

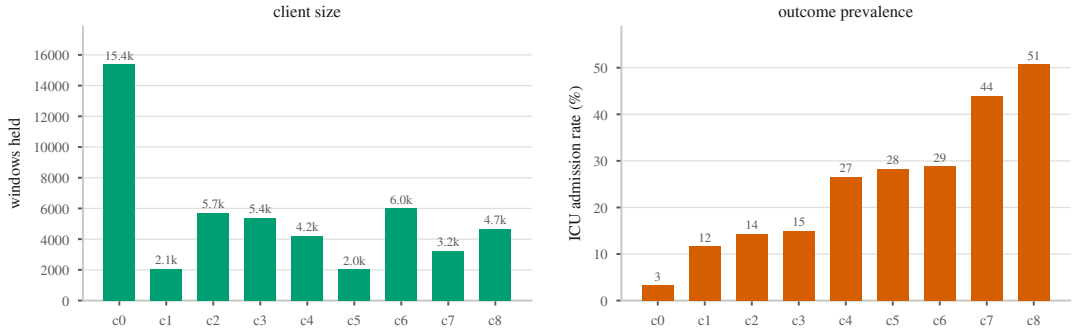



Fig. 6. Heterogeneity of the VitalDB clients (surgical specialties, case-level partition): size and intensive-care admission rate.

TABLE V
THE THREE CORPORA

| | VitalDB | WEAR | eICU |
|---|---|---|---|
| Clients | 9 specialties | 22 participants | 187 hospitals |
| Ind. | 5,974 | 22 | 132,736 |
| Windows (k) | 33.5/6.8/6.9 | 26.4/5.4/6.0 | 2122/564/570 |
| $\bar{k}$ | 7.9 | ≈1,750 | 28.5 |
| Modality A | 18-ch. trends | 12-ch. inertial | 12-ch. vitals |
| Modality B | 34 preop. | 2048 video | 5 admission |
| Task | ICU admission | 18 activities | mortality |
| Prevalence | 18.84 % | – | 4.22 % |

Ind.: individuals; $\bar{k}$: mean records per individual; windows are training/validation/test, in thousands. VitalDB: person-level partition (the case-level partition has 6,257 cases, 34.0/7.3/7.4k windows and a prevalence of 19.65 %).

TABLE VI
TRAINING AND EVALUATION PARAMETERS

| Parameter | Value |
|---|---|
| Cloud rounds $R$; edge rounds $\tau_e$ | 10; 5 |
| Edge aggregators $E$ | 2 (eICU scaling test: 2–16) |
| Local passes per edge round | 1 |
| Optimizer | Adam |
| Learning rate; weight decay | $10^{-3}$; $10^{-4}$ |
| Expected individuals per step $B$ | 64 |
| Clipping norm $C$ | 1 |
| Training budget | $\varepsilon = 0.9$, $\delta = 10^{-5}$ |
| Accountant | privacy loss distribution, per-client $\sigma_k$ |
| Records per patient (cap) | 64 |
| Update encoding | top-20 %, 8-bit, error feedback |
| Seeds | 5 (paired); 3 for grids |
| LiRA shadow models | 16 |
| Latency timing | P95, 1,000 runs, 100 warm-up |
| Software | PyTorch 2.5, Opacus 1.5, ONNX Runtime, ns-3 3.46, CloudSim Plus 8.5 |
| Hardware | one NVIDIA RTX A2000 GPU |

187 clients with a median of 444 individuals.

On eICU and WEAR, splits are made by individual. VitalDB is evaluated on two partitions. The *person-level* partition assigns each person to one split and one client; cases that would place a person in a second split or on a second client are removed (200 cases), leaving 4,221 training, 868 validation and 885 test persons. The *case-level* partition of the earlier experiments splits by operation, so that 55 of 940 test persons also have a training case and 68 training persons span two clients; results on it protect the case rather than the person and are marked case-level. The privacy cost, the cost of the unit, the comparison with prior systems and the membership-inference audit are person-level. All partitions are fixed, and no test case of either was used to pretrain the encoders. Four controls prevent inflation that would not be visible during training: splits by case rather than by window, which would place near-duplicates in the test set; modality B restricted to preoperative variables, because intraoperative interventions are given in response to deterioration and encode the outcome; missing values filled within a case and paired with an explicit indicator, because, for example, an arterial line is placed only in patients judged at risk; and normalization statistics computed on training cases only, or on the disjoint pool for the end-to-end eICU run. For eICU, a pool of 526,867 windows from individuals disjoint from every client is reserved before federated training to pretrain the disjoint encoders.

### B. Implementation

Table VI lists the training and evaluation parameters. All arms share one configuration, so that differences can be attributed to the factor under test. The headline system uses per-client multipliers and the privacy loss distribution accountant (Section IV-C). On eICU every pretrained encoder is disjoint. On VitalDB and WEAR, results use the oracle encoders unless stated, so absolute utilities there are upper bounds, whereas differences between arms, which share the encoders, are unaffected. The trained model is exported to ONNX and timed with ONNX Runtime on the CPU provider of a workstation (Intel Core i7-14700), not on edge hardware; the network is simulated in ns-3 and energy is modeled in CloudSim Plus.

### C. Metrics and Statistics

The primary utility measure is the area under the ROC curve (AUROC); average precision (AUPRC) is reported because the clinical tasks are imbalanced. Accuracy at the default threshold is uninformative under this imbalance: a constant negative prediction attains the base rate. Balanced accuracy is therefore reported at a threshold chosen on the validation cohort by Youden's index. Case-level metrics average the predictions of

a case's windows. Comparisons between arms are paired by seed, and intervals are

$$\bar{d} \pm t_{S-1,\,0.975}\; s_d/\sqrt{S} \tag{9}$$

over $S$ seeds with per-seed differences $d_s$. An effect is called *resolved* when this interval excludes zero; an unresolved estimate is reported as such and is not read as evidence of absence. Grids run at three seeds estimate how a cost changes across a factor, not each cell individually. Seed counts were fixed before each comparison and were not increased after inspecting results. The reported checkpoint is the one with the best validation AUROC, and each is evaluated once on the test split.

### D. Experimental Protocol

*System objectives* compare PP-HFM with centralized training and with the hierarchy alone. *Privacy cost* (O2b) compares the private system with an identical non-private system over five paired seeds; the primary measure (AUROC) and the secondary measures (AUPRC, balanced accuracy) were declared before the runs. *Recipe decomposition* uses three arms at $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$: B4, full-model private training from random initialization (prior art); B7, full-model private training from the pretrained initialization; and B6, head-only private training from the pretrained initialization (PP-HFM). B4→B7 changes only the initialization and B7→B6 only the privatized subset. *Records per individual*: the number of training records retained per individual, $k$, is varied with the number of units per step held equal in both arms, which removes the sampling-rate effect and leaves only the clipping geometry; validation and test sets are unchanged, and $k = 1$ is the control of Corollary 1. *Ablations* remove one component at a time. *Fusion* is measured outside the federated system, each arm retrained on its own inputs, because zeroing a stream of a jointly trained model measures robustness rather than information.

## VI. Results

### A. System Objectives

Table VII reports every objective of Section III on the three deployments, and Fig. 7 shows the system costs behind O1, O1b and O4. All targets are met on VitalDB. On WEAR all are met except the utility bound, which is inconclusive. On eICU five of seven are met; the utility bound is inconclusive under the prespecified interval criterion, with a point estimate above it, and is missed end to end; update delivery falls short at two edges.

*Communication and latency.* Two-tier aggregation of the compressed head update reduces uplink traffic by 90.8, 96.1 and 92.3 %. The measurement includes the index array that sparsification transmits with the values, which analytical estimates omit. Round latency falls from 90.5 to 75.2 ms on VitalDB, from 104.5 to 81.8 ms on WEAR and from 261.4 to 174.1 ms on eICU, and the reduction is positive in every simulated round. It is smaller than the traffic reduction because the hierarchy shortens link time, not computation, and it grows with the number of stations sharing a medium. Measuring every round at packet level matters: an earlier configuration that composed nine rounds in ten analytically placed the communication term of the centralized baseline at 256 ms against 23.6 ms simulated, overstating the benefit fourfold.

TABLE VII
Objectives, measured or simulated, against the targets fixed in advance

| | Quantity | VitalDB | WEAR | eICU | Target |
|---|---|---|---|---|---|
| O1 | Uplink reduction | 90.8% | 96.1% | 92.3% | $\geq 50\%$ |
| O1b | Round latency cut | 16.9% | 21.7% | 33.4% | $> 0\%$ |
| O2a | Budget $\varepsilon$ | 0.900 | 0.766[§] | 0.900 | $\leq 1$ |
| O2b | Utility loss [pts] | $-1.18$ | 1.97 pp[†§] | 3.10[†] | $\leq 2$ |
| O3a | P95 latency [ms] | 2.30 | 0.75 | 0.24 | $< 100$ |
| O3b | Update delivery | 100.00% | 99.94% | 97.83%* | $> 99\%$ |
| O4 | Energy saving | 72.3% | 76.8% | 79.6% | $\geq 20\%$ |

Measured: inference latency on a workstation CPU (ONNX Runtime) and the uplink payload of the deployed update encoding. Simulated: delivery and round latency (packet-level ns-3, every round) and energy (CloudSim Plus model). O2b: paired loss against an identical non-private system, five seeds, in AUROC points. [§]Record-level; O2b is the accuracy loss over fifteen seeds. [†]Interval spans the threshold (eICU: $[1.55, 4.65]$, point estimate above it; end to end 3.88 $[2.60, 5.17]$, not met). *Not met. VitalDB: O2a and O2b person-level, other rows case-level. eICU: O2a and O2b with each person at one hospital, other rows on the source partition (Section IV-B). O2a: training budget; the evaluated, non-deployed conformal release adds $\varepsilon_{\text{conf}} = 0.1$ on VitalDB and eICU (total 1.0) and 0.2 on WEAR (total 0.966).

TABLE VIII
Inference latency on VitalDB (1,000 timed runs, workstation CPU)

| Stage | Mean [ms] | P95 [ms] |
|---|---|---|
| Preprocessing | 0.106 | 0.110 |
| Forward pass | 2.130 | 2.165 |
| Conformal set (not deployed) | 0.019 | 0.021 |
| Post-processing | 0.007 | 0.008 |
| Total | 2.263 | 2.299 |
| Total, eight-bit graph | 3.515 | 17.337 |

After 100 warm-up runs. The eight-bit graph was evaluated and not deployed.

*Inference latency.* Timed on a workstation CPU (Intel Core i7-14700), not on edge hardware, the 95th-percentile latency is 2.30, 0.75 and 0.24 ms against a budget of 100 ms (Table VIII). The private mechanism adds no inference cost, because its noise acts only on training gradients; eight-bit quantization was rejected because it raised the 95th percentile to 17.34 ms.

*Reliability and energy.* Update delivery is 100.00 and 99.94 % on VitalDB and WEAR. On eICU, 187 clients on two edges contend for the shared medium, and delivery is 97.83 %, against 95.11 % for centralized training. With the edge count varied and everything else fixed, delivery rises monotonically to 99.94 % at sixteen edges, meets the target on average from six edges and in every round from eight (about 23 clients per edge, minimum 99.49 %). Energy per round falls by 72.3, 76.8 and 79.6 %. Introducing the edge tier alone increases total energy, from 1,266 to 1,344 J on VitalDB; the saving comes from the reduced device computation and transmission of PP-HFM, which lowers the device term fivefold.

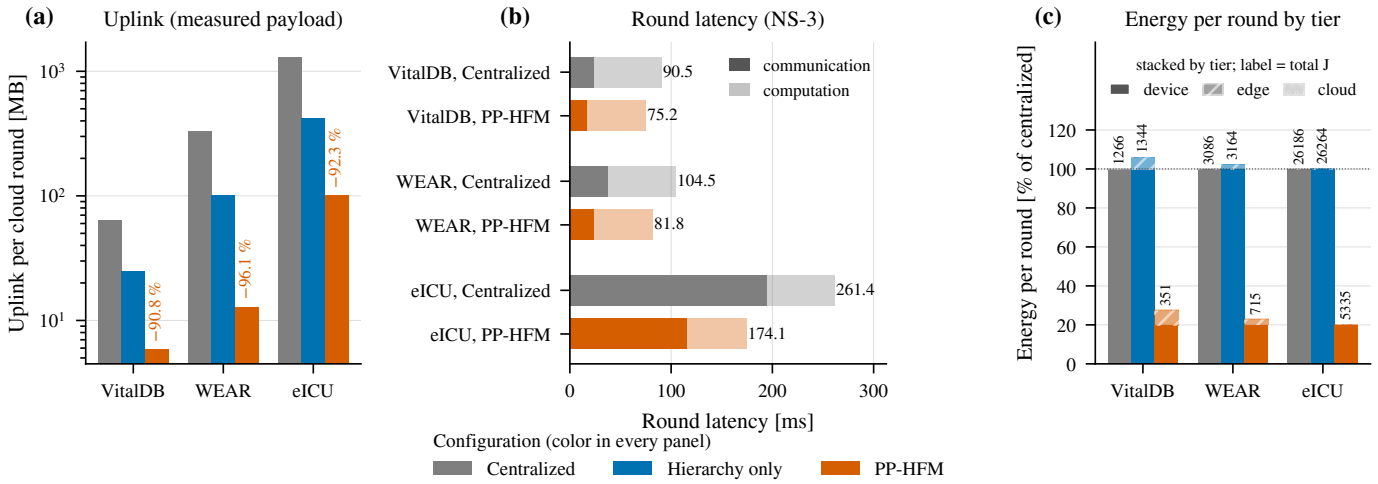



Fig. 7. System cost: (a) uplink per cloud round, (b) round latency, (c) energy per round by tier, relative to centralized training.

### B. Privacy Budget and Utility

On both clinical corpora, the contribution to the training transcript of every training patient is protected at $\varepsilon = 0.900$, $\delta = 10^{-5}$, across all tiers; on eICU these runs use the partition in which each person is held by one hospital (Section IV-B). This is not an end-to-end guarantee for the headline runs: their standardization constants are fitted on the training cohort, and on VitalDB and WEAR the oracle encoders also violate (A2) (Table III). The VitalDB and WEAR results are therefore comparisons of the private head, not end-to-end private deployments; the only configuration that satisfies the guarantee end to end is the eICU rerun with disjoint encoders and pool-fitted constants, reported separately below. The per-client multipliers range from 6.34 to 15.56 on VitalDB and from 4.85 to 29.34 on eICU (median 13.76); a single worst-case multiplier would have imposed 29.34 on every hospital.

*A private calibration threshold.* A per-client conformal threshold released by the exponential mechanism over ranks [31], [32] ($\varepsilon_{\text{conf}} = 0.1$) was evaluated and is not part of the system: with about a hundred calibration patients per client the private threshold returns the majority-class set on every VitalDB seed and on four of five eICU seeds, so coverage equals the base rate (80.1 % on VitalDB, case-level partition; 95.8 % on eICU) and misses the 90 % target. The non-private threshold does not collapse (positive-class coverage 36–43 %), but its coverage, 80.4–81.8 %, also misses the target.

At the default threshold the private model predicts the majority class on every VitalDB seed and on four of five eICU seeds (accuracy 0.8113 and 0.958, the base rates), so accuracy measures the threshold rather than the model, and the declared measures are used. On VitalDB the private model attains an AUROC of 0.8318 against 0.8200 for the non-private reference, a change of $-1.18$ points $[-1.63, -0.73]$ ($p = 0.002$); AUPRC changes by $-3.47$ $[-6.33, -0.61]$ and balanced accuracy by $-0.31$ $[-1.23, +0.62]$, negative values favoring the private model. O2b is met. The gain is consistent with regularization of the compact head by clipping and noise, which the case-level ablation suggests (Section VI-F) but this experiment does not isolate; on the case-level partition the change was $-0.90$ $[-1.82, +0.02]$. On eICU a cost is resolved: AUROC falls from 0.8239 to 0.7929, by 3.10 points $[1.55, 4.65]$; the interval spans the two-point target, so O2b is inconclusive under the prespecified interval criterion at five seeds, although the point estimate exceeds the target. AUPRC falls by 1.84 $[1.60, 2.08]$, spanning the threshold, and balanced accuracy by 0.33 $[-0.22, +0.88]$, which meets it. The cost is set by the smallest hospitals, which sample every patient at every step and carry the largest multipliers. With standardization constants fitted on the disjoint pool and encoders pretrained in those units, so that the guarantee holds end to end, the eICU cost is 3.88 points $[2.60, 5.17]$, 2.31 $[1.42, 3.21]$ in AUPRC and 2.80 $[2.52, 3.08]$ in balanced accuracy: O2b is missed in AUROC and in balanced accuracy, whose intervals lie above the target, while the AUPRC interval spans it with a point estimate above it. Assigning each person to one hospital left the cost of privacy unchanged within resolution: against the same arms on the source partition, paired by seed, the AUROC cost differs by $+0.01$ $[-0.72, +0.75]$ points with training-fitted and by $-0.03$ $[-0.66, +0.60]$ with pool-fitted constants; on the source partition the end-to-end balanced-accuracy interval, $[1.57, 3.82]$, had spanned the target. On WEAR, whose guarantee is record-level, the accuracy loss is 1.97 pp $[1.76, 2.17]$, spanning the threshold.

*Cost of the unit.* Moving the guarantee from the record to the patient at a matched budget ($\varepsilon = 1$, five seeds, VitalDB) changes AUROC by $-0.0170$ $[-0.0269, -0.0071]$ ($0.8423 \rightarrow 0.8254$) and AUPRC by $-0.0582$ $[-0.0701, -0.0462]$ (case-level: $-0.0147$ and $-0.0783$). Two effects contribute: sampling patients rather than records reduces the sampled units from 33,491 to 4,221, and clipping a patient's summed gradient constrains individuals with many records most. Section VI-D isolates the second.

*Cost of disjoint encoders.* With the disjoint instead of the oracle encoders, the VitalDB model (case-level partition) loses $1.57 \pm 0.46$ AUROC points ($0.8395 \rightarrow 0.8237$, five paired seeds, $p < 0.001$) and $1.22 \pm 2.62$ AUPRC points, which is not resolved. Against a non-private reference with the same disjoint encoders, the cost of privacy is $+0.14$ $[-0.33, +0.62]$

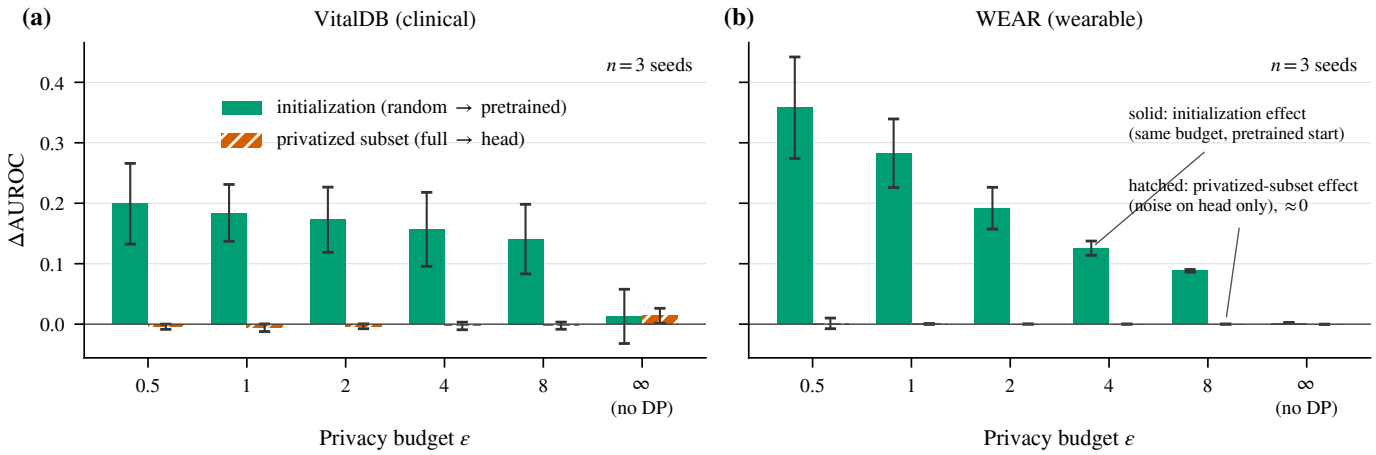



Fig. 8. Recipe decomposition across $\varepsilon$ (AUROC, $n = 3$, 95 % CI; VitalDB case-level): solid, initialization effect; hatched, privatized-subset effect.

TABLE IX
RECIPE DECOMPOSITION IN AUROC (THREE SEEDS; VITALDB CASE-LEVEL)

| | VitalDB | | WEAR | |
|---|---|---|---|---|
| $\varepsilon$ | B4→B7 | B7→B6 | B4→B7 | B7→B6 |
| $\infty$ | $+0.0129$ | $+0.0139 \pm 0.0124$ | $+0.0021$ | $-0.0002 \pm 0.0003$ |
| 8 | $+0.1407$ | $-0.0025 \pm 0.0059$ | $+0.0883$ | $+0.0001 \pm 0.0002$ |
| 4 | $+0.1569$ | $-0.0029 \pm 0.0063$ | $+0.1257$ | $+0.0001 \pm 0.0002$ |
| 2 | $+0.1727$ | $-0.0035 \pm 0.0040$ | $+0.1918$ | $+0.0002 \pm 0.0001$ |
| 1 | $+0.1841$ | $-0.0059 \pm 0.0063$ | $+0.2827$ | $+0.0003 \pm 0.0008$ |
| 0.5 | $+0.1991$ | $-0.0042 \pm 0.0043$ | $+0.3580$ | $+0.0013 \pm 0.0088$ |

B4→B7 changes only the initialization; B7→B6 only the privatized subset (whole network → head). Mean ± Student-$t$ half-width. VitalDB uses the patient-level mechanism on the case-level partition; WEAR is record-level.

TABLE X
AUPRC GAP BETWEEN THE RECORD AND PATIENT ARMS (THREE SEEDS; VITALDB CASE-LEVEL)

| | VitalDB ($\varepsilon_{\text{tr}} = 1$) | | eICU ($\varepsilon_{\text{tr}} = 0.8$) | |
|---|---|---|---|---|
| $k$ | Gap | $p$ | Gap | Forecast |
| 1 | $-0.0071 \pm 0.0599$ | 0.661 | $-0.0011 \pm 0.0369$ | 0.0000 |
| 2 | $+0.0486 \pm 0.0129$ | 0.004 | $+0.0048 \pm 0.0057$ | 0.0515 |
| 4 | $+0.0796 \pm 0.0163$ | 0.002 | $+0.0081 \pm 0.0554$ | 0.0816 |
| 8 | $+0.0872 \pm 0.0248$ | 0.004 | $+0.0099 \pm 0.0437$ | 0.0876 |
| 16 | – | – | $+0.0103 \pm 0.0332$ | 0.0878 |

Record arm minus patient arm, paired by seed; mean ± Student-$t$ half-width. The forecast for eICU was registered before the corpus was built.

AUROC points, which meets O2b, and $+2.21$ $[-0.03, +4.46]$ AUPRC points. This configuration satisfies (A2), but its standardization constants are still fitted on the training individuals; on eICU, where the disjoint pool is not carved from a small cohort, every result already uses disjoint encoders.

### C. Initialization, Not the Privatized Subset

PP-HFM differs from conventional private training in two respects: it starts from pretrained encoders and privatizes only the head. Table IX and Fig. 8 separate the two. On VitalDB (case-level partition), under the patient-level mechanism, changing only the initialization is worth $0.0129$ AUROC without noise and $0.1991$ at $\varepsilon = 0.5$, increasing monotonically as the budget tightens; on WEAR it ranges from $0.0021$ to $0.3580$. Under privacy, restricting the privatized subset changes AUROC by between $-0.0059$ and $-0.0025$ on VitalDB and by at most $0.0013$ on WEAR, with intervals covering zero at every finite budget. Only without noise does the head-only arm score higher on VitalDB, by $0.0139 \pm 0.0124$; no privacy mechanism is involved there.

The result contradicts a dimensionality explanation. If a smaller privatized vector absorbed less noise, B7, which privatizes all $1{,}864{,}227$ parameters, should trail B6, which privatizes $559{,}874$, most at the tightest budget. At $\varepsilon = 0.5$ the two differ by $0.0042$, within the interval and with B7 the higher, consistent with the noise scale being set by the budget, the step count and the sampling rate rather than by the number of parameters. Across the budget range, the randomly initialized arm loses $0.1906$ on VitalDB and $0.3716$ on WEAR, where it falls to $0.6247$, close to chance; the pretrained full-model arm loses $0.0043$ and the head-only arm $0.0225$ on VitalDB, and both less than $0.016$ on WEAR. Restricting the subset is therefore justified by communication, since only the head is transmitted, not by accuracy.

### D. Records per Individual

Table X and Fig. 9 report the gap between the record- and patient-level arms as the number of training records per individual $k$ varies, with the units per step equal in both arms (VitalDB case-level).

*VitalDB.* At $k = 1$ the arms are indistinguishable ($-0.0071$, interval covering zero), as Corollary 1 requires, so the divergence at larger $k$ is attributable to the unit. From $k = 2$ the gap is resolved and grows to $0.0872$ at $k = 8$. The patient-level arm is flat (means $0.5083$–$0.5128$), whereas the record-level arm rises from $0.5015$ to $0.5955$ (Fig. 10): the widening gap is additional records that the record-level arm exploits and the patient-level arm cannot, because the bound on the summed contribution is tightest for the individuals who contribute them.

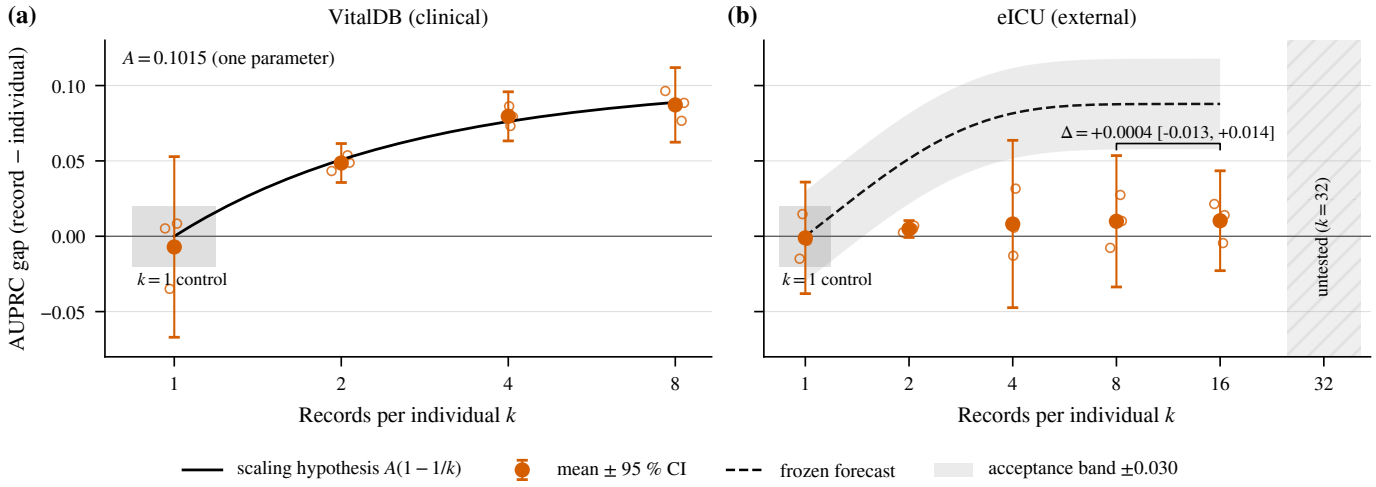



Fig. 9. Patient-level cost (AUPRC gap) against records per individual $k$, $n = 3$: (a) VitalDB (case-level) with the fitted curve $A(1 - 1/k)$; (b) eICU against the frozen forecast.

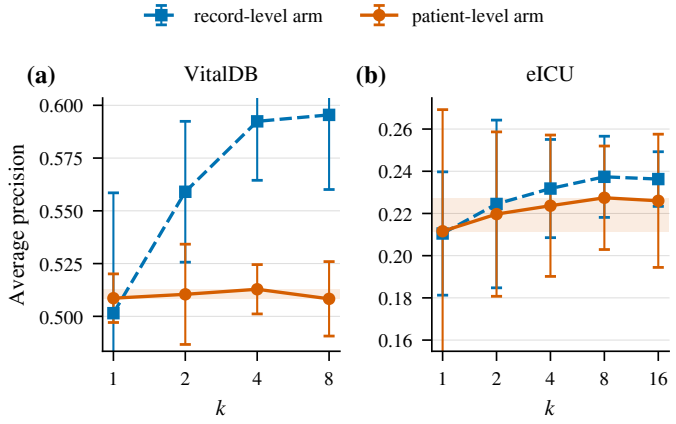



Fig. 10. AUPRC of each arm against $k$ ($n = 3$, 95 % CI; VitalDB case-level): the patient-level arm is flat; the record-level arm improves.

AUROC moves in the same direction (+0.0026 to +0.0159) but is unresolved at three seeds.

*Test of the scaling hypothesis.* The one-parameter curve $A(1-1/k)$, fitted to the twelve paired gaps, gives $A = 0.1015$ at a root-mean-square error of 0.0119, against 0.0117 for a two-parameter saturating curve $A(1 - e^{-b(k-1)})$: the extra parameter buys nothing. Fitted on $k \leq 4$ only, it predicts 0.0904 at $k = 8$ against 0.0872 observed. The patient-level arm changes by less than 0.001 per doubling of $k$, as the proposition requires.

*External test on eICU.* Before the eICU corpus was built, a forecast was registered from the VitalDB grid ($A = 0.0878$, $b = 0.883$) with an acceptance band of ±0.030. On eICU the $k = 1$ control holds (−0.0011), and the ordering and the saturation reproduce: the gap rises through $k = 8$ and changes by +0.0004 [−0.0134, +0.0141] from $k = 8$ to $k = 16$, meeting the registered bound on further growth. The magnitude does not: at $k = 8$ the gap is 0.0099, about one ninth of the forecast, and no eICU gap is resolved at three seeds. Attainable precision differs by a factor of two between

TABLE XI
AUPRC GAP ACROSS THE THREE SCALING GRIDS (THREE SEEDS; VITALDB CASE-LEVEL)

| | | VitalDB | | eICU |
|---|---|---|---|---|
| $k$ | Forecast | $\varepsilon_{\mathrm{tr}} = 1.0$ | $\varepsilon_{\mathrm{tr}} = 0.8$ | $\varepsilon_{\mathrm{tr}} = 0.8$ |
| 1 | 0.0000 | −0.0071 | +0.0398 | −0.0011 |
| 2 | 0.0515 | +0.0486 | +0.0364 | +0.0048 |
| 4 | 0.0816 | +0.0796 | +0.0981 | +0.0081 |
| 8 | 0.0876 | +0.0872 | +0.0914 | +0.0099 |

The two VitalDB grids differ only in the operating point. The forecast was fitted on VitalDB at $\varepsilon_{\mathrm{tr}} = 1$ and never refitted.

the tasks (AUPRC near 0.22 at a prevalence of 4.2 %, against 0.5), and a cost that is a fixed fraction of attainable precision would preserve the shape at a lower ceiling; distinguishing this from a corpus-specific ceiling requires a third corpus. The registered design extended to $k = 32$, but only 20.2 % of eICU individuals reach 32 records, all of them readmitted, so the maximum was reduced to 16 by a dated amendment before any result beyond $k = 8$ was computed.

*Budget confound.* The forecast was fitted at $\varepsilon = 1$ and tested at 0.8 with a batch of 64. Re-running VitalDB at the eICU operating point (Table XI) does not shrink its gaps (+0.0914 against +0.0872 at $k = 8$; the one-parameter curve fitted on $k \leq 4$ predicts 0.0988), so the budget does not explain the shortfall. That grid, however, fails its own $k = 1$ control (+0.0398, all seeds of one sign), which combines the smallest training pool with the largest batch; subtracting each grid's $k = 1$ offset attributes about half (51 %) of the difference at $k = 8$ to the operating point, and under neither reading does the corpus term change sign.

*Clipping rule.* The flatness of the patient-level arm could stem from the clipping rule rather than the unit. Normalizing each patient's summed gradient by its record count, or its square root, before clipping leaves the sensitivity and the budget unchanged and tests this. On the natural eICU partition, where records per individual range from 2 to 192,

TABLE XII
FUSION ON EICU AS THE STRONGER STREAM IS WEAKENED (AUROC, FIVE SEEDS; GAINS OVER THE BETTER SINGLE MODALITY).

| Condition | Vitals | Admission | Fused | Gain | Gain (AP) |
|---|---|---|---|---|---|
| Matched control | 0.7068 | 0.8221 | 0.8241 | +0.0020 | −0.0008 |
| Feature zeroed | 0.6973 | 0.5947 | 0.6911 | −0.0062 | −0.0182 |
| Titrated to parity | 0.6992 | 0.6845 | 0.7282 | **+0.0291** | +0.0121 |

the count normalization lowers the final-round clipped fraction from 0.14 to 0.08 but changes AUPRC by $-0.0012$ $[-0.0093, +0.0069]$ (three seeds); the one-sided 95 % upper bound, $+0.0043$, excludes the pre-registered recovery of $+0.010$, and square-root normalization changes AUPRC by $-0.0031$. On VitalDB at $k = 8$ and the same budget, count normalization lowers the final-round clipped fraction from 0.50 to 0.34 and changes AUPRC by $+0.0038$ $[-0.0072, +0.0148]$; the one-sided upper bound, $+0.0113$, does not exclude $+0.010$, so this test is inconclusive by its own criterion, and square-root normalization changes AUPRC by less than $10^{-4}$. The clipping rule therefore does not explain the cost on eICU, and on VitalDB it explains at most one eighth of the 0.0914 cost at that operating point.

### E. Multimodal Fusion

On VitalDB (case-level partition) the fused model attains $0.8653 \pm 0.0072$ AUROC against $0.8283 \pm 0.0087$ for the better single modality, a gain of 0.0370 at 5.9 seed standard deviations, and 0.0671 in AUPRC (five seeds). On WEAR the better single modality already reaches 0.9976, leaving no headroom. On eICU fusion yields nothing in the natural configuration ($+0.0037$, and $+0.0020$ on the partitions of the test below), where the admission stream (0.8221) dominates the vital-sign stream (0.7068). A pre-registered test weakened the stronger stream to parity by mixing its dominant feature with variance-preserving noise, calibrated on that stream alone and never on the fused model (Table XII, Fig. 11). At parity the fused model exceeds the better single modality by $0.0291 \pm 0.0211$, positive on all five seeds ($p = 0.019$), fourteen times the matched control; the AUPRC gain, $0.0121 \pm 0.0215$, is not resolved. Balance is therefore sufficient for a fusion gain on this corpus, but not shown to be necessary, and the balanced model remains below the intact admission stream. Because the eICU second stream is admission data rather than a second sensor, this controlled test identifies a mechanism; it does not show that a two-sensor clinical deployment would gain as much.

### F. Component Ablation and Membership-Inference Audit

Each component is removed in turn from the full system, three seeds paired (case-level) (Table XIII, Fig. 12(a)). Removing sparsification ($+0.0012$) or quantization ($+0.0001$, the narrowest interval in the study) leaves accuracy unchanged and raises uplink fivefold and 1.6-fold, so compression is justified by traffic alone. Removing the edge tier costs $-0.0058$ at three seeds and is resolved at the six pre-registered seeds, $-0.0047$ $[-0.0088, -0.0006]$ on VitalDB and $-0.0052$

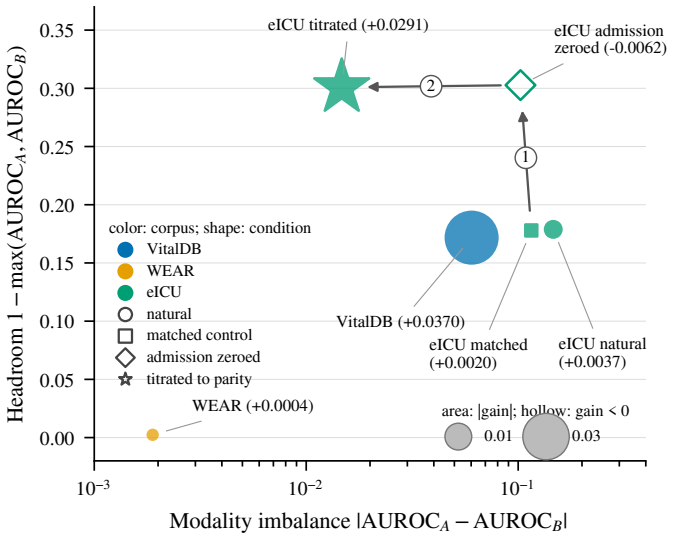



Fig. 11. Fusion gain against modality imbalance and headroom (color: corpus; shape: condition; area: |gain|); numbered arrows trace the eICU titration.

TABLE XIII
COMPONENT ABLATION ON VITALDB (RECORD-LEVEL MECHANISM, CASE-LEVEL PARTITION, THREE SEEDS)

| Removed | AUROC | ΔAUROC | MB |
|---|---|---|---|
| — (full) | 0.8541 | — | 252 |
| C1 hierarchy | 0.8484 | $-0.0058 \pm 0.0088$ | 50 |
| C2 privacy | 0.8307 | $\mathbf{-0.0234 \pm 0.0205}$ | 252 |
| C3 sparsification | 0.8553 | $+0.0012 \pm 0.0083$ | 1260 |
| C4 quantization | 0.8543 | $+0.0001 \pm 0.0006$ | 403 |
| +C6 loss weighting | 0.8542 | $+0.0001 \pm 0.0009$ | 252 |

ΔAUROC is paired against the full system; bold: the interval excludes zero. MB: total uplink. C6 is an inclusion arm.

$[-0.0074, -0.0030]$ on WEAR; because the flat arm also changes the aggregation schedule, this bounds the cost of the hierarchy rather than claiming a benefit. Removing the privacy mechanism *reduces* AUROC by 0.0234: two of three non-private seeds peak in the first round and then degrade, whereas every private seed peaks between rounds six and nine, so clipping and noise regularize a compact head over frozen features. The loss-aware edge weighting of an earlier version (C6) adds nothing and is not adopted.

*Audit.* Table XIV reports the leakage of three configurations on the person-level partition. The positive control, non-private training of the same head, shows that the audit detects leakage: offline LiRA reaches 0.605, 0.052 above its difficulty floor and 0.048 above the loss threshold, and online LiRA with a global variance reaches 0.597 $[0.586, 0.609]$, entirely above the floor; at three seeds the offline interval spans the floor. Record-level training remains detectably above its floor under offline LiRA (0.576, margin $+0.065$), whereas patient-level training does not (0.503, margin $-0.005$). The record-level arm is not equivalent to the control: it lies below it under the loss threshold (0.522 against 0.556) and online LiRA with a global variance (0.508 against 0.597), and the two were not compared statistically (three control seeds against five). The attack that validates the control (online, global variance) is

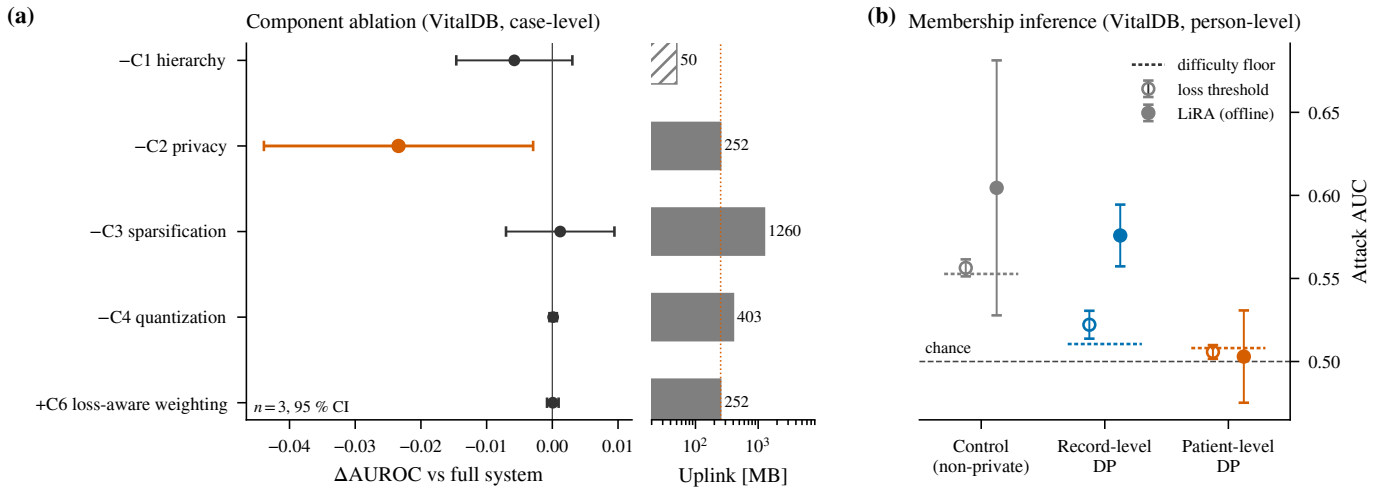



Fig. 12. VitalDB: (a) component ablation, case-level (ΔAUROC, $n = 3$, 95 % CI); (b) membership inference, person-level, against each difficulty floor (dotted).

TABLE XIV
MEMBERSHIP INFERENCE ON VITALDB (PERSON-LEVEL PARTITION)

| Configuration | Loss AUC | LiRA AUC | Floor |
|---|---|---|---|
| Control (no DP) | 0.556 [0.551, 0.561] | 0.605 [0.528, 0.681] | 0.553 |
| Record-level DP | 0.522 [0.514, 0.531] | 0.576 [0.557, 0.594] | 0.511 |
| Patient-level DP | 0.506 [0.502, 0.510] | 0.503 [0.475, 0.531] | 0.508 |

Mean attack AUC with 95 % interval over the target seeds (three for the control, five otherwise); LiRA is the offline likelihood-ratio attack. Floor: the AUC from per-patient difficulty alone, without the target model.

thus not the one that separates the two units (offline). Paired over five seeds on identical shadow splits, the record-level arm exceeds the patient-level arm by $+0.073$ $[0.054, 0.092]$ ($p < 0.001$) for offline LiRA and by $+0.016$ $[0.010, 0.023]$ ($p = 0.002$) for the loss threshold, both surviving correction for four attacks; the two online attacks resolve no difference. The direction is as predicted, but the magnitude is smaller: the record-level attack reaches 0.576, not the predicted 0.7. On the case-level partition, where members were operations, no difference was resolved ($+0.029$ $[-0.038, +0.095]$ for offline LiRA). The audit covers VitalDB only and bounds what these attacks achieve; it does not establish the absence of leakage.

*Attribution.* Integrated gradients over 512 test windows (case-level) give the 22 missingness indicators 15.8 % of the prediction without privacy and 21.6–21.7 % under either unit with frozen encoders, below their 42 % share of inputs.

### G. Comparison with Prior Systems

HED-FL [3] and FL2DP [9] were re-implemented and trained through the same trainer, splits, frozen encoders, rounds and five seeds as PP-HFM (Table XV). HED-FL aggregates hierarchically without privacy, uploads dense updates and samples 60 % of the clients per round; FL2DP perturbs every coordinate with an exponential mechanism and shuffles the updates, without a hierarchy. FL2DP was run on VitalDB only, because its per-coordinate mechanism would take about two GPU days per seed on eICU.

*Utility and traffic.* On VitalDB, PP-HFM exceeds FL2DP by 0.0123 AUROC $[0.0064, 0.0181]$ and 0.025 AUPRC $[0.015, 0.035]$, and HED-FL by 0.030 AUPRC $[0.005, 0.055]$; the AUROC difference from HED-FL, 0.0056 $[-0.0018, +0.0131]$, is not resolved. HED-FL exceeds the non-private PP-HFM reference by 0.0062 AUROC $[0.0011, 0.0113]$. On eICU, HED-FL is 3.17 AUROC points $[1.56, 4.77]$ and 1.77 AUPRC points $[0.75, 2.78]$ above the private model and level with the non-private reference ($+0.0008$, $p = 0.39$), so the gap is the price of the patient-level guarantee measured in Section VI-B. PP-HFM uploads 2.2 times less than HED-FL on VitalDB, 2.4 times less on eICU, and 4.0 times less than FL2DP.

*Leakage.* On VitalDB, under the audit of Section VI-F, on the same seeds and shadow splits, offline LiRA reaches 0.581 against HED-FL and 0.608 against FL2DP, 0.078 $[0.050, 0.106]$ and 0.105 $[0.079, 0.130]$ above PP-HFM; the loss threshold gives $+0.047$ and $+0.055$. Both baselines also have higher difficulty floors (0.555 and 0.559 against 0.508), and their margins above them are $+0.026$ and $+0.049$, against $-0.005$ for PP-HFM. The PP-HFM is the patient-level audit target, trained at $\varepsilon = 1.0$ with $\varepsilon_{\text{conf}} = 0.5$ and one worst-client multiplier under the Rényi accountant, not the $\varepsilon = 0.9$ model whose AUROC and uplink are listed, so its leakage is measured at a slightly weaker budget than the headline. The comparison bounds what these attacks achieve and does not by itself establish the guarantee. FL2DP leaks more than HED-FL, which adds no noise at all. Its candidate set is built from the private gradient value (Eqs. 11–12 of [9]), so the mechanism satisfies no finite $\varepsilon$, and the audit finds that it does not reduce leakage either.

## VII. DISCUSSION

### A. What the Price of the Unit Depends On

The central finding is not that an individual-level guarantee has a cost, which was expected, but that the cost has a predictable shape. At one operating point on person-level VitalDB it is 0.0170 AUROC. Holding the per-step budget and

TABLE XV
PP-HFM AGAINST RE-IMPLEMENTED PRIOR SYSTEMS (FIVE SEEDS)

| | System | $\varepsilon$ | AUROC | MB | LiRA |
|---|---|---|---|---|---|
| VitalDB | PP-HFM | 0.90 | **0.8318** | **252** | **0.503** ($-0.005$) |
| | HED-FL | $\infty$ | 0.8262 | 560 | 0.581 ($+0.026$) |
| | FL2DP | — | 0.8195 | 1008 | 0.608 ($+0.049$) |
| eICU | PP-HFM | 0.90 | 0.7936 | **5235** | — |
| | HED-FL | $\infty$ | **0.8252** | 12541 | — |

$\varepsilon$: patient-level training budget ($\delta = 10^{-5}$); —: no finite bound. MB: total uplink. LiRA: offline attack AUC, margin above the difficulty floor in parentheses; not run on eICU. The PP-HFM LiRA entry is the patient-level audit target (Section IV-F: $\varepsilon = 1.0$, $\varepsilon_{\text{conf}} = 0.5$, one worst-client multiplier, Rényi accountant), not the $\varepsilon = 0.9$ model whose AUROC and uplink are listed. eICU rows: source partition, on which the baselines were run (2.9 % of training persons at several hospitals, Section IV-B). All AUROC differences from PP-HFM are resolved except HED-FL on VitalDB.

the units per step fixed, so that only the clipping geometry differs, it is indistinguishable from zero at one record per individual and grows as $A(1 - 1/k)$, with a single parameter that a fit on $k \leq 4$ already determines. The patient-level arm stays flat while the record-level arm improves, so a patient-level guarantee becomes relatively more expensive, not cheaper, as more records are collected per patient. Normalizing the clipping bound by the record count does not recover the cost on eICU and could explain at most one eighth of it on VitalDB, so the cost belongs mainly to the unit. Proposition 1 also points to what would: clipping around an estimate of each individual's mean gradient, whose user-level rates improve with $k$ [21]. The external corpus reproduces the shape and the saturation but not the magnitude, so the shape is interpreted as structural and the magnitude as corpus-dependent.

### B. Position Against Prior Art

PP-HFM builds on user-level privacy [17] and parameter-efficient private fine-tuning [15], [16]; its contribution is their composition inside a hierarchy, with a proof, and its cost. Hierarchical systems [3], [7], [8] report savings without a guarantee, and private systems [9], [10] leave the edge tier out of the accounting; PP-HFM measures both on the same configuration and shows that the savings cost no utility. The re-implemented baselines make this concrete (Section VI-G). On eICU, HED-FL matches the non-private reference, so its 3.2-point lead over PP-HFM is the price of the patient-level guarantee; on VitalDB, PP-HFM matches HED-FL in AUROC, exceeds it in AUPRC, and leaks less under offline LiRA (margin $-0.005$ against $+0.026$). FL2DP adds noise yet leaks more than HED-FL, which adds none: a perturbation without a finite $\varepsilon$ is not a weaker guarantee but no guarantee, and only an audit can show what it protects. The parameter-efficient literature attributes robustness at small $\varepsilon$ to the reduced privatized dimension; the three-arm decomposition attributes it to the initialization instead, which changes what a practitioner should optimize first. Several results are negative and are reported with the control or the power that makes them interpretable: the privatized subset contributes nothing to accuracy once the initialization is controlled, compression and loss-aware weighting do not affect accuracy, and count-normalized clipping does not recover the cost of the unit on eICU.

### C. Practical Implications

Four recommendations follow. First, choose the unit explicitly and price it in records per individual: where individuals contribute many records, a record-level budget bounds the wrong quantity. Second, secure a legitimate pretrained initialization before tuning anything else; under a tight budget it is the largest effect measured here. Third, calibrate each client to its own population (on eICU a single worst-case multiplier would impose 29.34 on every hospital against a median need of 13.76), and consider a minimum cohort size, since the smallest hospitals set the cost. Fourth, use compression and the edge tier, which cut traffic and energy at no measurable cost in accuracy, and establish fusion per task.

### D. Limitations

*Utility objective.* On eICU the guarantee costs 3.10 AUROC points $[1.55, 4.65]$, inconclusive against the two-point target under the prespecified interval criterion although the point estimate exceeds it, and 3.88 $[2.60, 5.17]$ end to end, above the target; on WEAR the accuracy interval spans the target. *Encoders.* VitalDB and WEAR results use the oracle encoders unless stated, so their absolute utilities are upper bounds; on the case-level VitalDB partition the disjoint encoders cost 1.57 AUROC points and the privacy cost with them is $+0.14$ points; differences between arms are unaffected, and on eICU every encoder is disjoint. *Scope of the unit.* VitalDB results other than the privacy cost, the cost of the unit, the baseline comparison and the audit are case-level; for the 3.9 % of persons with several operations, their guarantee extends to the person only by group privacy. On WEAR the mechanism is record-level. The validation cohort is used without noise for checkpoint selection. *Statistical power.* The scaling grids and the recipe sweep use three seeds; on eICU no gap of Table X is resolved, and resolving it would need about twenty-five. The audit (sixteen shadows, 600 non-members, VitalDB only; about 160 GPU hours on eICU) cannot resolve a 0.1 % false-positive rate, and at three seeds its positive control is resolved by online LiRA only, which does not separate the two units; the audit bounds these attacks and does not establish the absence of leakage. *Baselines.* HED-FL and FL2DP are re-implemented from their papers, and FL2DP and both baseline audits are limited to VitalDB. *Simulation.* Network terms are packet-level simulated and compute terms modeled, so round latencies compare topologies rather than measure a deployed system. *Fusion.* A fusion gain is shown on VitalDB only: WEAR leaves no headroom, and the eICU second stream is admission data, so its balancing test identifies a mechanism rather than the benefit of a second sensor. *Theory.* The $A(1-1/k)$ form assumes both arms attain their rate with a common constant; it is supported empirically, not proved.

## VIII. Conclusion

This paper addressed a mismatch between what private hierarchical federated learning guarantees and what clinical

IoT requires: existing systems protect individual records, whereas a clinical deployment must protect patients, each of whom contributes many records. The proposed framework, PP-HFM, closes this gap by construction. Each client sums a patient's gradients before clipping, calibrates its noise to its own population, which parallel composition permits when every patient is held by one client, and accounts for every release of the hierarchy, edge tier included. The guarantee was proved for the training transcript; it holds end to end only when the encoders and standardization constants are fitted on public data, which on the evaluated corpora is the eICU configuration, run on a partition in which every individual is held by one hospital.

The evaluation on three corpora supports four conclusions. First, the system objectives are met on every corpus with two exceptions on eICU and one on WEAR, where the record-level utility loss of $1.97$ pp is inconclusive against the two-point target: the training budget is at most $\varepsilon = 0.9$ ($1.0$ with the evaluated conformal release), uplink traffic falls by $90.8$–$96.1\,\%$, modeled energy per round by $72.3$–$79.6\,\%$, and the $95$th-percentile inference latency on a workstation CPU is at most $2.3$ ms, but on eICU the utility loss exceeds the two-point target end to end and is inconclusive against it with training-fitted constants, and update delivery reaches $99\,\%$ only from eight edges. Second, the patient-level guarantee is affordable: it costs no utility on VitalDB, partitioned by person, and on eICU it costs $3.1$ AUROC points $[1.6, 4.7]$ ($3.9$ $[2.6, 5.2]$ end to end), a cost set by the smallest hospitals. Third, on VitalDB and WEAR, robustness to a strict budget is determined by the initialization, not by the size of the privatized subset: pretrained initialization is worth $0.199$ AUROC at $\varepsilon = 0.5$, whereas restricting noise to the head changes accuracy by at most $0.006$ under privacy and is justified by communication alone. Fourth, the price of an individual-level guarantee follows $A(1-1/k)$ in records per individual and vanishes where the two units coincide, a scaling that holds out of sample on the first corpus and in shape on the second. On VitalDB, a patient-level audit with a positive control is consistent with the distinction: record-level training remains detectably above its difficulty floor under offline LiRA, whereas patient-level training does not; the audit bounds these attacks without establishing the absence of leakage. The corpora carry different evidence: eICU tests the guarantee end to end, VitalDB tests it on the training transcript, and WEAR, where each client is one participant, evaluates the system at record level.

Several directions remain open. First, a third corpus is needed to determine whether the ceiling of the cost curve is a constant of the guarantee or a fraction of the precision attainable on a task. Second, the noise imposed by the smallest clients could be reduced, for example by sampling patients across the clients of an edge. Third, clipping around an estimate of each individual's mean gradient, which the analysis indicates could recover the cost of the unit, should be tested. Finally, the audit should be extended to eICU and, with a larger shadow population, to low false-positive rates.

## Data Availability

VitalDB [27] (https://vitaldb.net) and WEAR [28] are publicly available. The eICU Collaborative Research Database [36] is available through PhysioNet [37] to credentialed users under a data use agreement, so no patient-level records or arrays derived from them are redistributed. All corpora are de-identified; this study is a secondary analysis with no patient contact. Code and aggregate results will be released upon publication.

## Acknowledgment

The authors thank Mr. Tejas J (Capgemini) for industry guidance throughout this work. They also thank the custodians of VitalDB (Seoul National University Hospital), of WEAR, and of the eICU Collaborative Research Database (the MIT Laboratory for Computational Physiology and the Philips eICU Research Institute) for making the data available.

## References

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.

[2] L. Liu, J. Zhang, S. H. Song, and K. B. Letaief, "Client-edge-cloud hierarchical federated learning," in *Proc. IEEE Int. Conf. Communications (ICC)*, 2020.

[3] F. De Rango, A. Guerrieri, P. Raimondo, and G. Spezzano, "HED-FL: A hierarchical, energy efficient, and dynamic approach for edge federated learning," *Pervasive and Mobile Computing*, vol. 92, p. 101804, 2023.

[4] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *Proc. IEEE Symp. Security and Privacy (S&P)*, 2017, pp. 3–18.

[5] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS)*, 2016, pp. 308–318.

[6] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.

[7] Z. Zhang, Z. Gao, Y. Guo, and Y. Gong, "Heterogeneity-aware cooperative federated edge learning with adaptive computation and communication compression," *IEEE Transactions on Mobile Computing*, vol. 24, no. 3, pp. 2073–2084, 2025.

[8] N. Mhaisen, A. A. Abdellatif, A. Mohamed, A. Erbad, and M. Guizani, "Optimal user-edge assignment in hierarchical federated learning based on statistical properties and network topology constraints," *IEEE Trans. Network Science and Engineering*, vol. 9, no. 1, pp. 55–66, 2022.

[9] C. Gu, X. Cui, X. Zhu, and D. Hu, "FL2DP: Privacy-preserving federated learning via differential privacy for artificial IoT," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 4, pp. 5100–5111, 2024.

[10] H. Zhou, Y. Zheng, and X. Jia, "Towards robust and privacy-preserving federated learning in edge computing," *Computer Networks*, vol. 243, p. 110321, 2024.

[11] S. Bao, S. Wu, B. Tanane, and Y. Tao, "Privacy-preserving communication compression and robust aggregation for hierarchical federated learning," *Complex & Intelligent Systems*, 2026.

[12] M. Rangwala, R. O. Sinnott, and R. Buyya, "Topology-aware differential privacy in hierarchical federated learning," arXiv:2506.19260, 2025.

[13] M. R. Hasan, M. I. Ahmed, S. Saha, T. K. Ishika, H. A. Shoaib, and M. J. Hossen, "Multi-modal federated learning with differential privacy for privacy-preserving healthcare AI," *Scientific Reports*, vol. 16, p. 21253, 2026.

[14] A. Siva M and C. L. Chowdhary, "Multi-modal personalized federated learning with adaptive differential privacy for medical image classification and a privacy-preserving approach," *Scientific Reports*, vol. 16, p. 21743, 2026.

[15] X. Li, F. Tramèr, P. Liang, and T. Hashimoto, "Large language models can be strong differentially private learners," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[16] D. Yu, S. Naik, A. Backurs, S. Gopi, H. A. Inan, G. Kamath, J. Kulkarni, Y. T. Lee, A. Manoel, L. Wutschitz, S. Yekhanin, and H. Zhang, "Differentially private fine-tuning of language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[17] H. B. McMahan, D. Ramage, K. Talwar, and L. Zhang, "Learning differentially private recurrent language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2018.

[18] I. Mironov, "Rényi differential privacy," in *Proc. IEEE Computer Security Foundations Symp. (CSF)*, 2017, pp. 263–275.

[19] I. Mironov, K. Talwar, and L. Zhang, "Rényi differential privacy of the sampled Gaussian mechanism," 2019, arXiv:1908.10530.

[20] S. Gopi, Y. T. Lee, and L. Wutschitz, "Numerical composition of differential privacy," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021, pp. 11 631–11 642.

[21] D. Levy, Z. Sun, K. Amin, S. Kale, A. Kulesza, M. Mohri, and A. T. Suresh, "Learning with user-level privacy," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[22] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *Proc. IEEE Computer Security Foundations Symp. (CSF)*, 2018, pp. 268–282.

[23] L. Watson, C. Guo, G. Cormode, and A. Sablayrolles, "On the importance of difficulty calibration in membership inference attacks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[24] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, "Membership inference attacks from first principles," in *Proc. IEEE Symp. Security and Privacy (S&P)*, 2022, pp. 1897–1914.

[25] N. Rieke, J. Hancox, W. Li, F. Milletarì, H. R. Roth, S. Albarqouni, S. Bakas, M. N. Galtier, B. A. Landman, K. Maier-Hein, S. Ourselin, M. Sheller, R. M. Summers, A. Trask, D. Xu, M. Baust, and M. J. Cardoso, "The future of digital health with federated learning," *npj Digital Medicine*, vol. 3, no. 119, 2020.

[26] M. J. Sheller, B. Edwards, G. A. Reina, J. Martin, S. Pati, A. Kotrotsou, M. Milchenko, W. Xu, D. Marcus, R. R. Colen, and S. Bakas, "Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data," *Scientific Reports*, vol. 10, no. 12598, 2020.

[27] H.-C. Lee, Y. Park, S. B. Yoon, S. M. Yang, D. Park, and C.-W. Jung, "VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients," *Scientific Data*, vol. 9, no. 279, 2022.

[28] M. Bock, H. Kuehne, K. Van Laerhoven, and M. Moeller, "WEAR: An outdoor sports dataset for wearable and egocentric activity recognition," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, no. 4, p. 175, 2024.

[29] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.

[30] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[31] A. Smith, "Privacy-preserving statistical estimation with optimal convergence rates," in *Proc. ACM Symp. Theory of Computing (STOC)*, 2011, pp. 813–822.

[32] F. McSherry and K. Talwar, "Mechanism design via differential privacy," in *Proc. IEEE Symp. Foundations of Computer Science (FOCS)*, 2007, pp. 94–103.

[33] J. Chen and X. Ran, "Deep learning with edge computing: A review," *Proceedings of the IEEE*, vol. 107, no. 8, pp. 1655–1674, 2019.

[34] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 2704–2713.

[35] R. Bassily, A. Smith, and A. Thakurta, "Private empirical risk minimization: Efficient algorithms and tight error bounds," in *Proc. IEEE Symp. Foundations of Computer Science (FOCS)*, 2014, pp. 464–473.

[36] T. J. Pollard, A. E. W. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, and O. Badawi, "The eICU Collaborative Research Database, a freely available multi-center database for critical care research," *Scientific Data*, vol. 5, no. 180178, 2018.

[37] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.